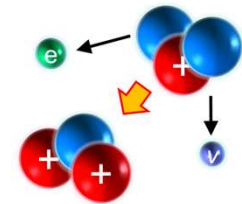


Essential Points of Interdisciplinary Engineering Sciences  
~ Material, **Energy**, Environment ~

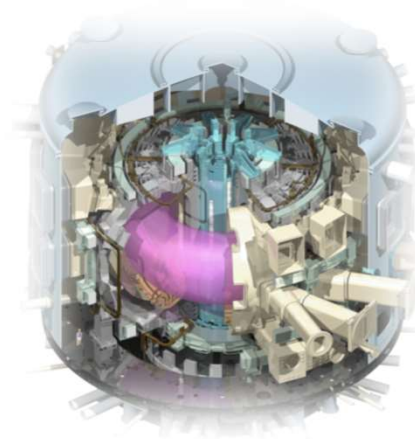
**Next Generation Energy source**

# **Nuclear Fusion Reactor**

~Tritium Science and Technology ~



Tritium



May 20<sup>th</sup>, 2025

<https://www.iter.org>



***Energy Chemical Engineering,***

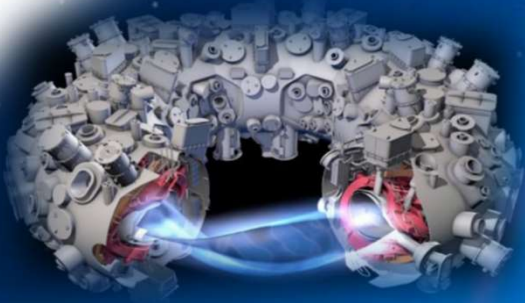
*Plasma and Quantum Science and Engineering major,*

*Interdisciplinary Graduate School of Engineering Sciences, Kyushu University*

***Fusion Energy Engineering***

**KATAYAMA Kazunari**

Nuclear fusion reactors are being developed as a next-generation energy source.



Helical Fusion Reactor



TOKAMAK Fusion reactor



Laser Fusion Reactor

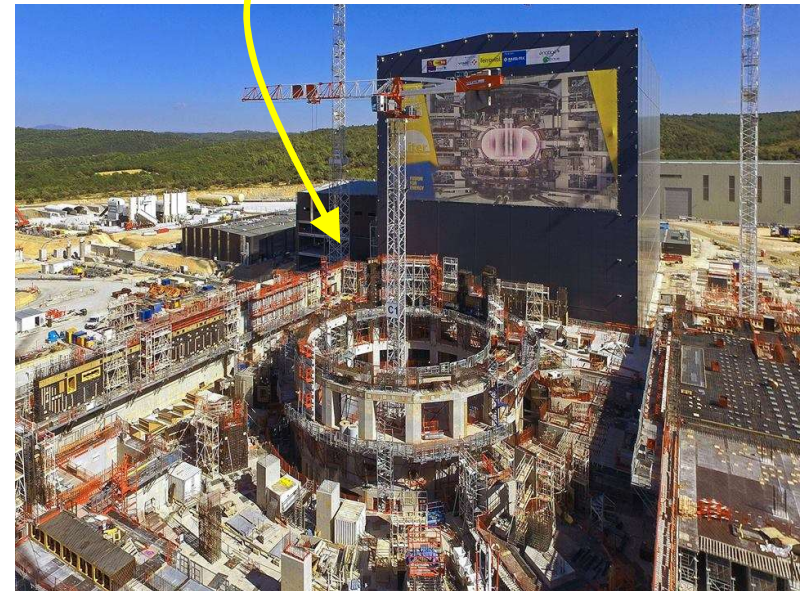
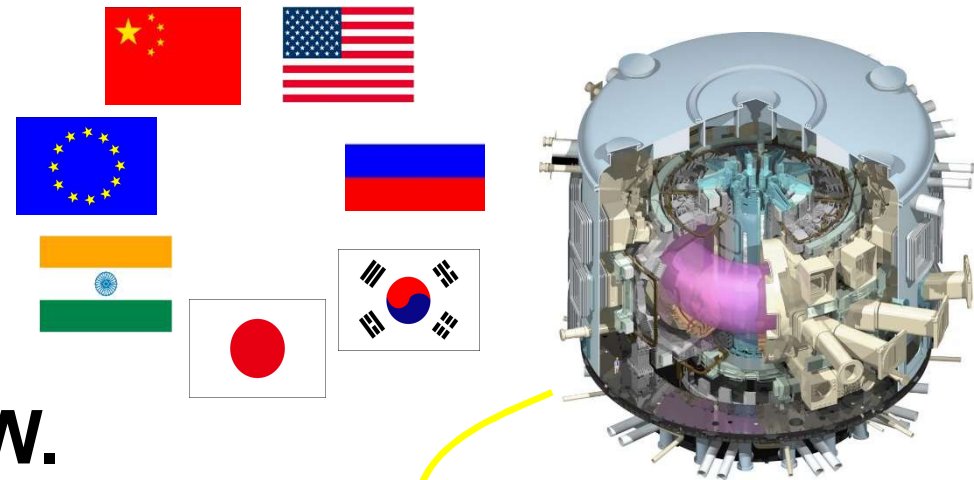
Here, the research activity for the fusion reactor engineering in Kyushu University will be introduced.

# ITER Project

**ITER** is constricting under the international collaboration.

**Thermal output is 500MW.**  
**Fuel cycle is operated.**

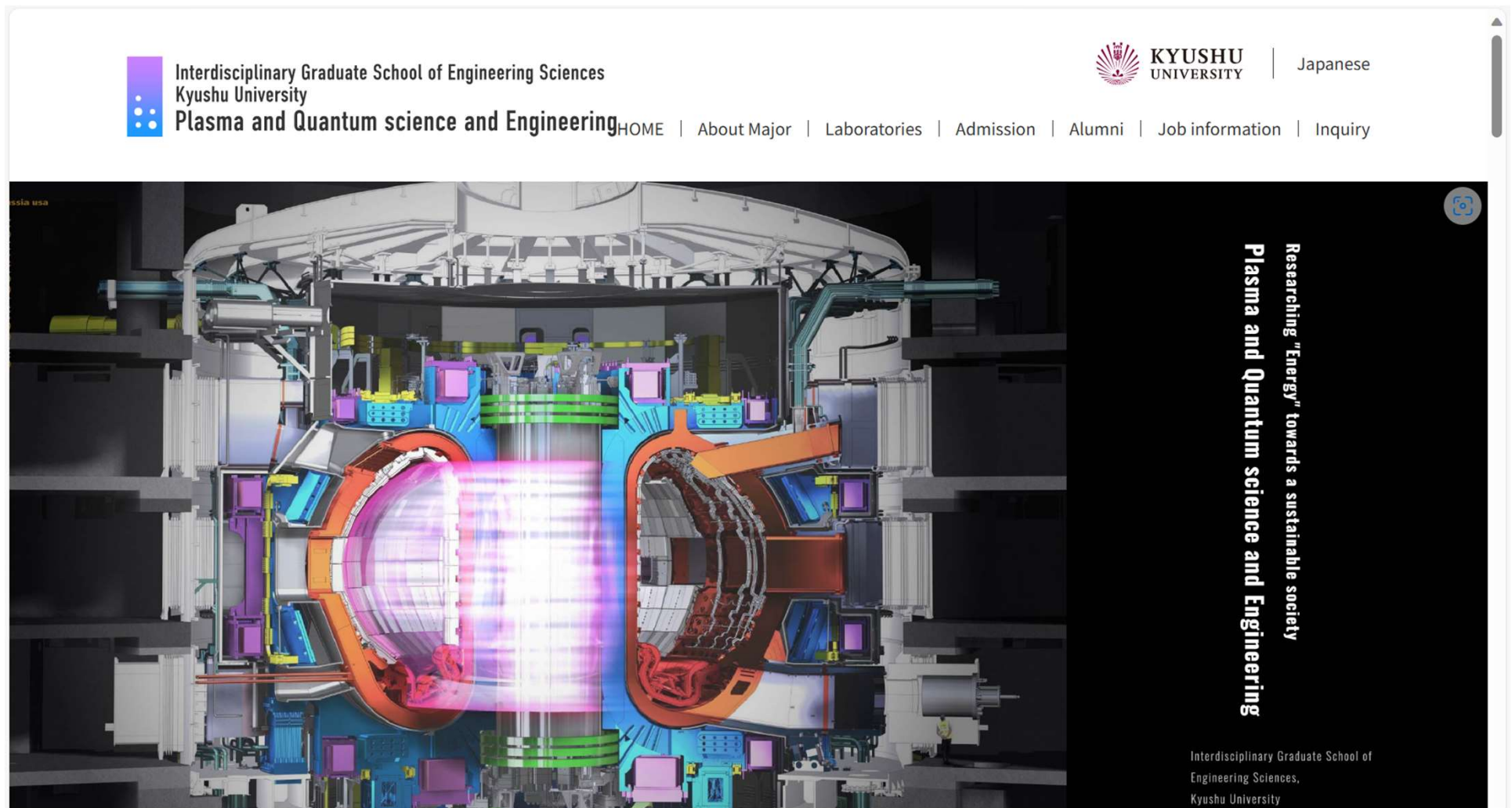
New energy for humanity of the future must be created by the energy of young scientists under international cooperation projects.



France, Cadarache



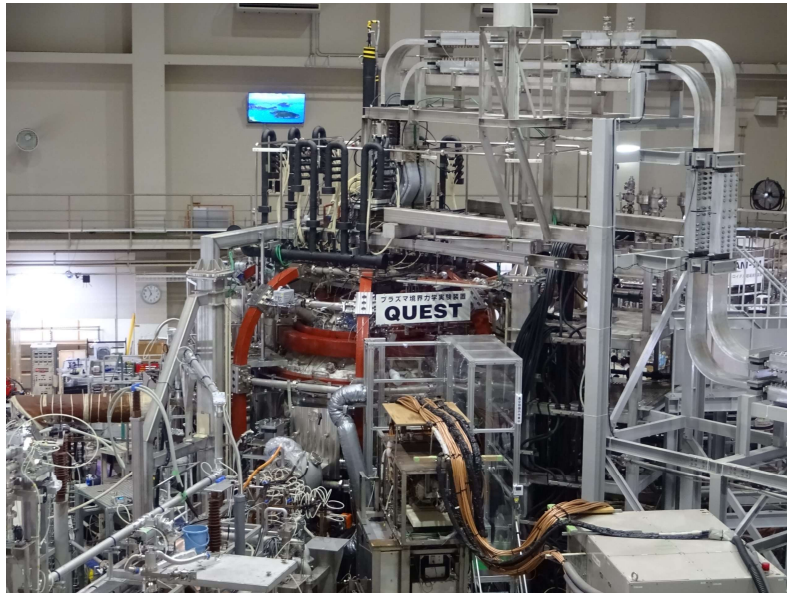
# Plasma and Quantum science and Engineering



ITER:Nuclear fusion reactor development project is progressing with international cooperation



# Research activities for Fusion in KU



## QUEST spherical tokamak

To maintain high-temperature plasma steadily.

To understand the physical properties of plasma through the development of measurement system



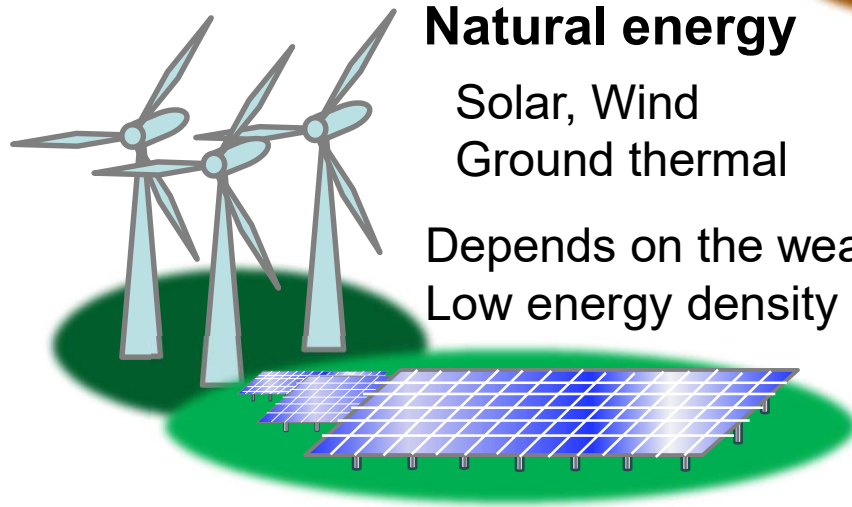
## Tritium handling room

To develop tritium safe handling technology

To understand tritium behavior in various materials and make adequate model of tritium behavior

# Main Energy Sources

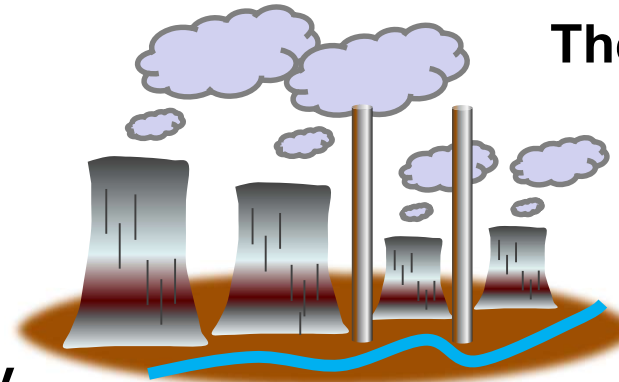
In order to maintain the current comfortable life, energy sources must be continuously stably operated.



## Natural energy

Solar, Wind  
Ground thermal

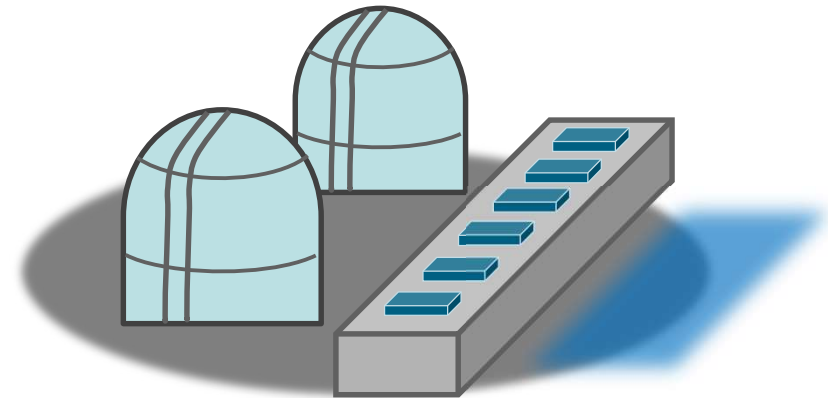
Depends on the weather.  
Low energy density



## Thermal power

Coal, Oil,  
Natural Gas

Limitation of fossil fuel reserves.  
Concerns about environmental impact

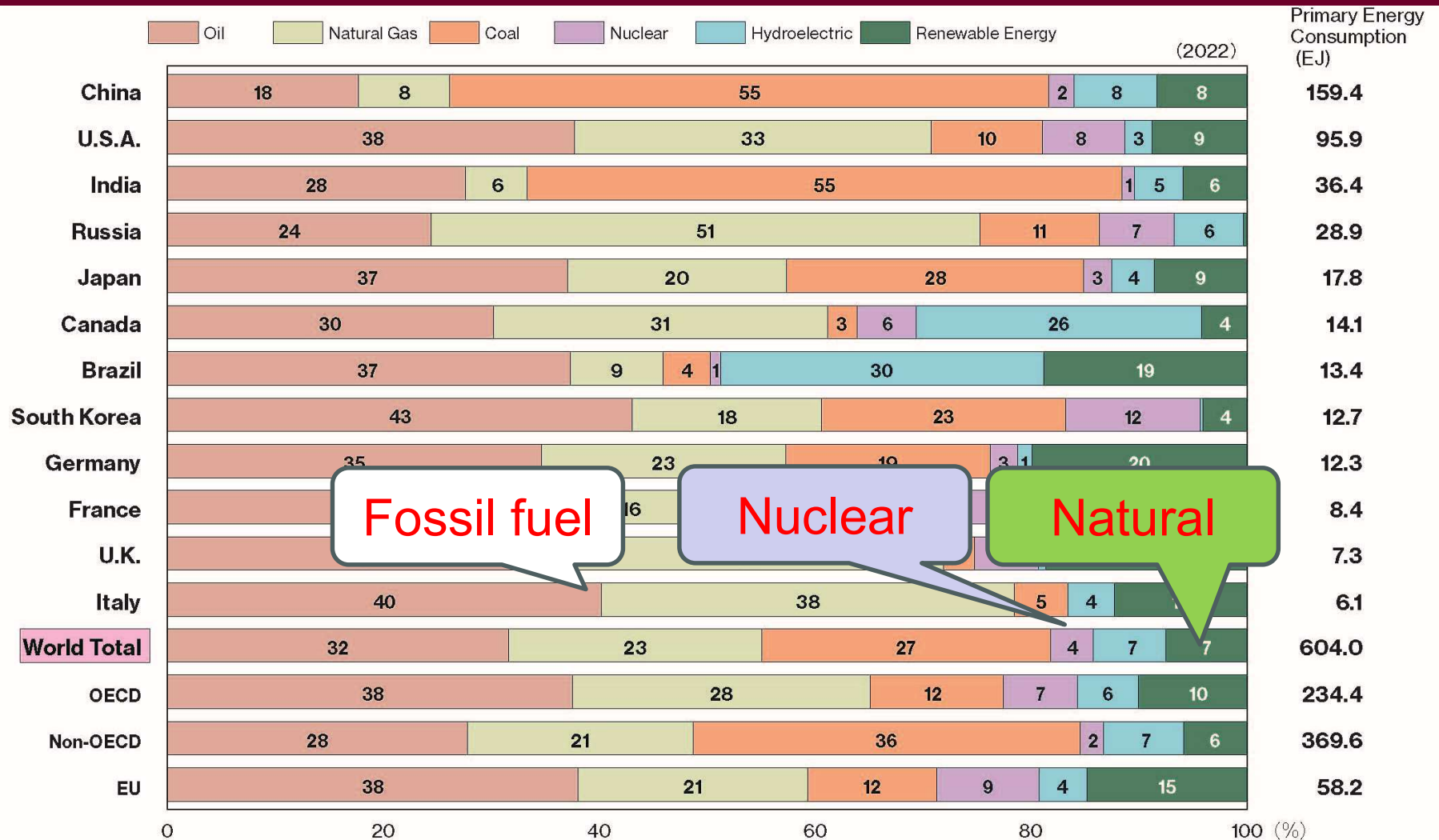


## Nuclear power (fission reactor)

Limitation of uranium reserves.  
Concerns about safety and RA wastes.

Each country will consider the best energy mix depending on the situation.

# Primary Energy Consumption in Major Countries



(Note) Figures may not add up to the totals due to rounding.

1 EJ (=10<sup>18</sup> Joules) is equivalent to the amount of heat from approximately 25,800,000 kℓ of crude oil (EJ: exajoule).

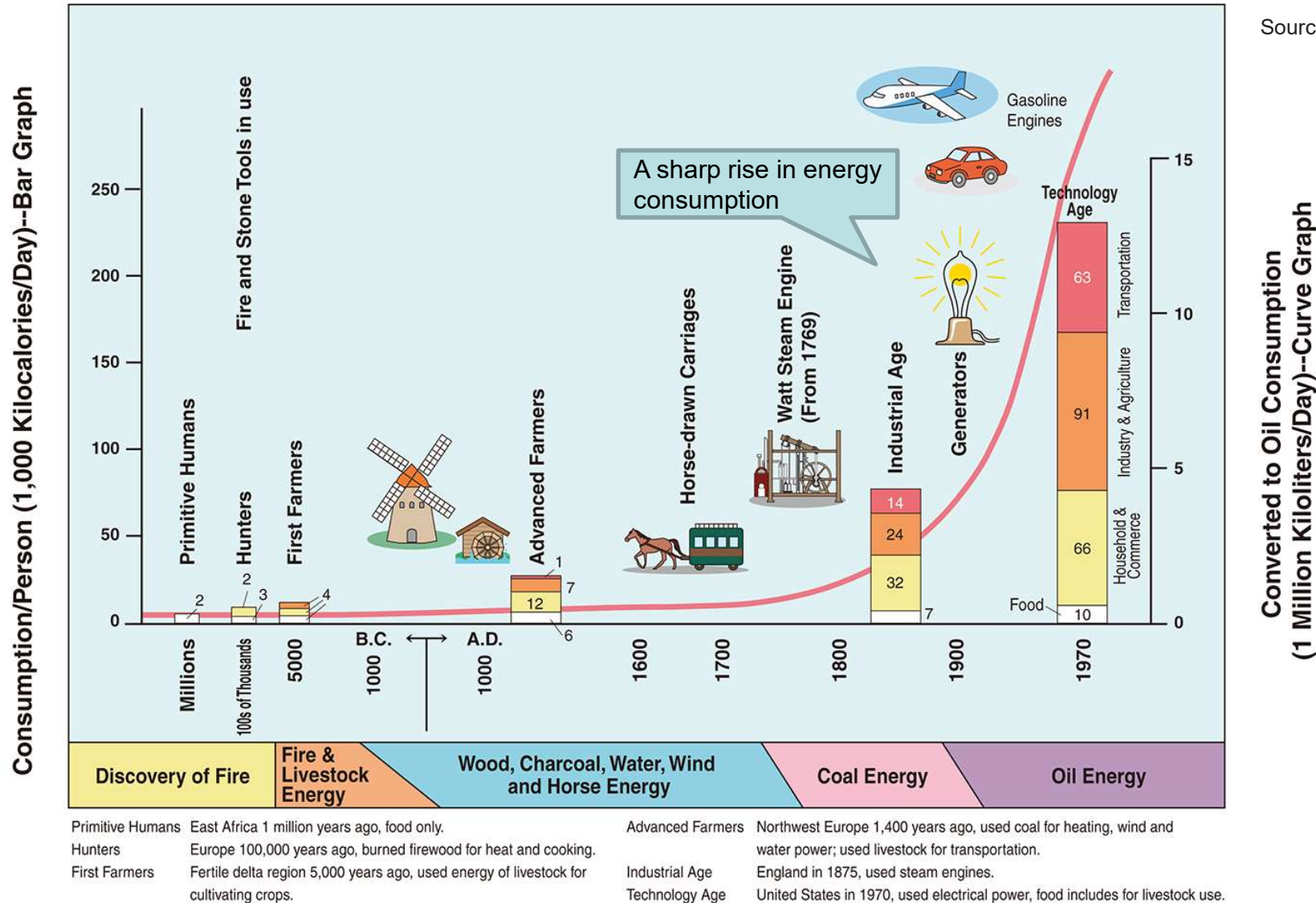
Source : JAERO (Japan Atomic Energy Relations Organization)

Our comfortable life is supported by fossil fuels that are easy to use and valuable energy sources.



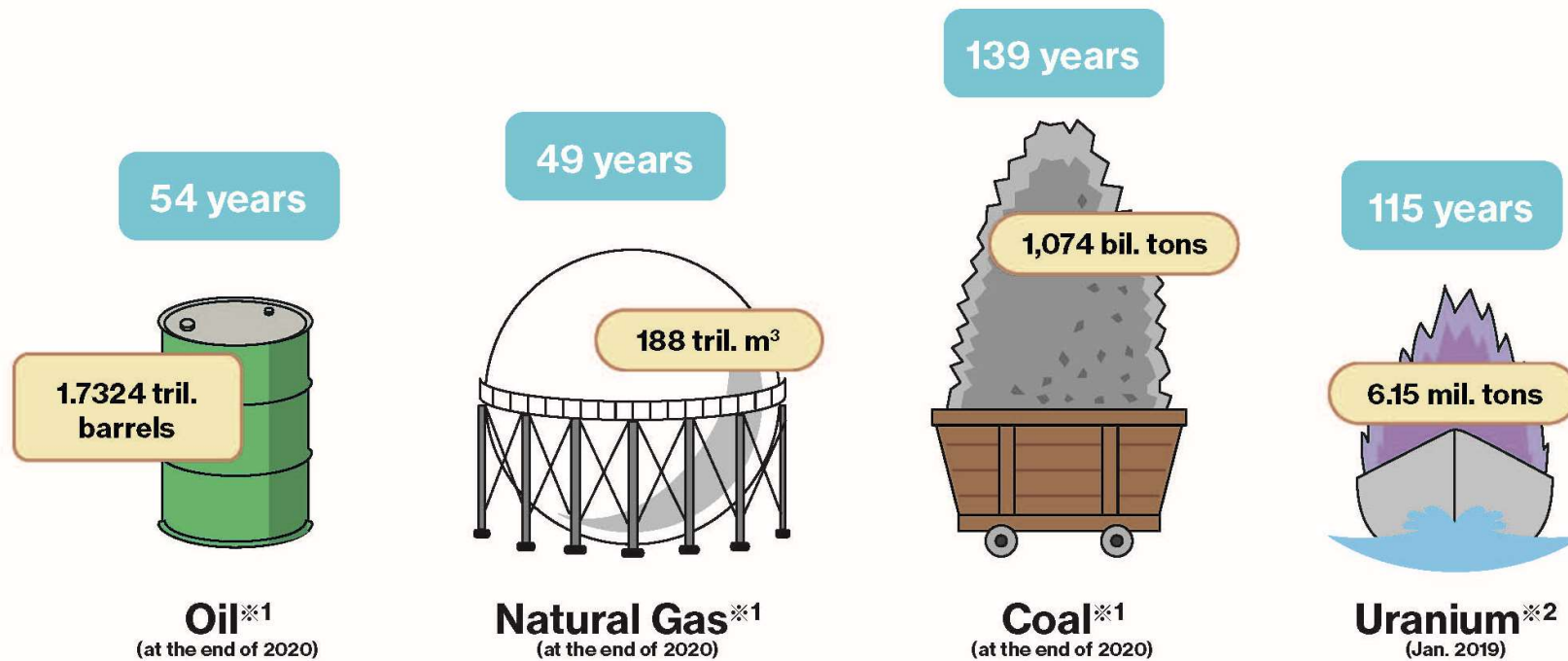
# Energy consumption and Age

Source : JAERO



Large amounts of fossil fuels have been consumed in the last 100 years. And it continues to increase.

# Proven Reserves of Energy Resources



(Note) Reserves-to-production (R/P) ratio = Proven Reserves / Annual Production

RAR (reasonably assured resources) of uranium is estimated at a production cost less than USD 130/kgU.

Source : JAERO

- ✓ Fossil fuels are a very convenient energy resource, but their reserves are limited.
- ✓ Nuclear power (fission) will continue to be used globally, although there are safety concerns. And, uranium reserves are also limited.
- ✓ It is necessary to save existing energy resources and leave them for future humanity.
- ✓ One promising solution is the realization of a fusion reactor.

# Requirements for energy sources

---

- Resources exist widely on the earth.  
For example, oil is concentrated in specific areas.
- Stable supply of sufficient energy.  
It is desirable to supply the required amount at the required time.
- Minimize environmental impact.  
CO<sub>2</sub> emission, acid rain etc.
- Safe and secure.  
Keep accidental risk as low as possible.
- The cost is within a reasonable range.  
We must consider the cost for the environment and the cost for decommissioning



# Features of Fusion reactors

---

- Resources exist widely on the earth.
  - Fuel resources (deuterium, lithium) are in the seawater, almost inexhaustible !!
- Stable supply of sufficient energy.
  - The energy released per unit mass of fuel is 3 times larger than that of a fission reactor !!
- Minimize environmental impact.
  - CO<sub>2</sub> emissions are very small.
  - No high-level long-lived radioactive waste !!

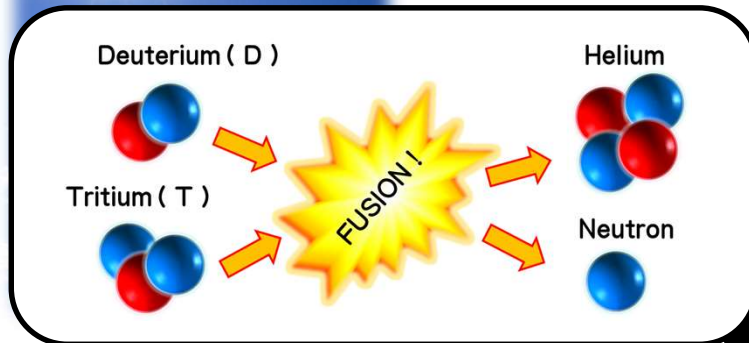
# Features of Fusion reactors

---

- Safe and secure.
  - No chain reaction in uncontrol, in principle safer than fission reactors !!
  - The decay heat density of the radioactive material is small, and then the requirement for heat removal is smaller than that of the fission furnace.
  - Tritium (T) management is the most important for safety.
    - ◆ The danger of T is sufficiently lower than cesium and iodine!
    - ◆ Since T is dispersed in plant, it will not be released at once !!
- The cost is within a reasonable range.
  - Evaluated to be about twice that of a coal plant.

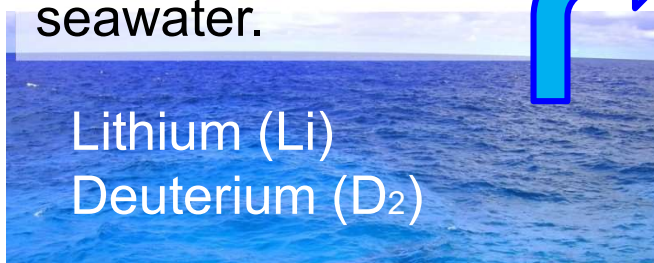
# Fusion Energy on the earth

The sunshine originates from nuclear fusion reaction, which generates enormous energy and is essential for life on the earth.

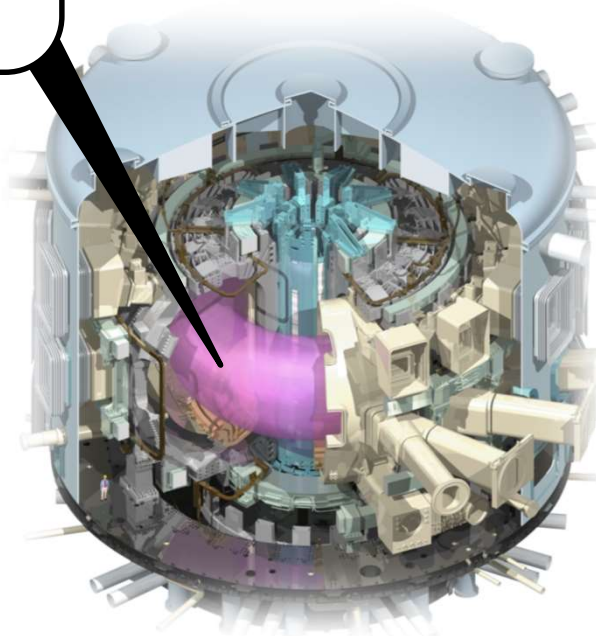


A fusion reactor is a device that controls the fusion reaction on the earth and tries to extract energy.

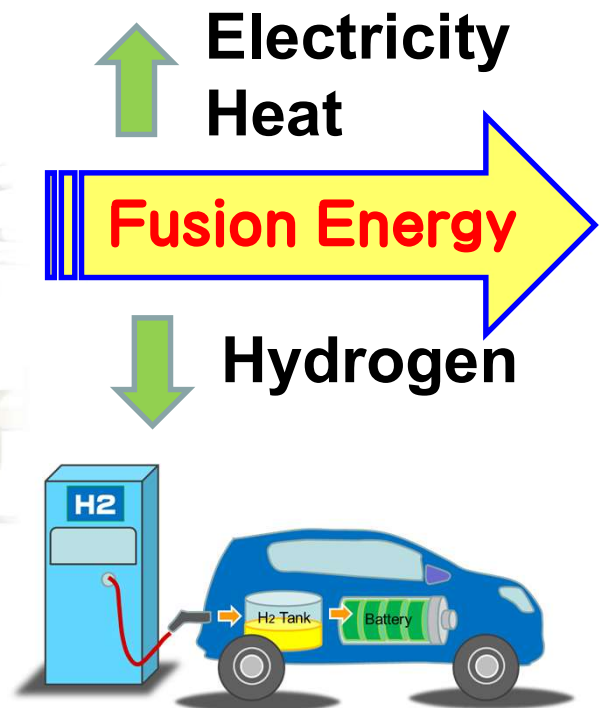
Fuel resource can be acquired from seawater.



Resources are almost inexhaustible.

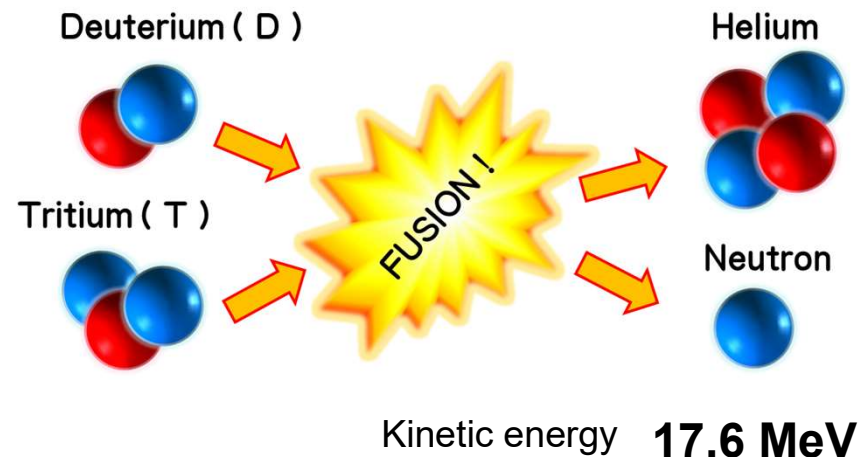
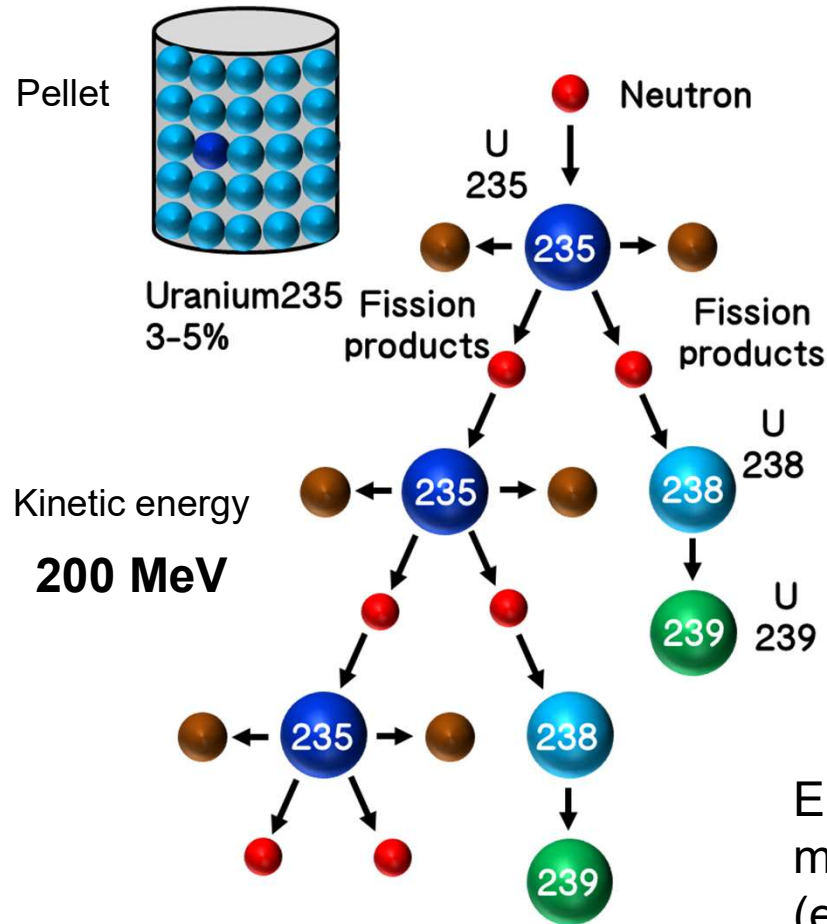


Fusion reactor



# Fission and Fusion

**Nuclear Fission reaction:** The process of generating heat by splitting heavy atoms. The splitting of atoms is achieved by hitting the heavy atom with high-speed particles, usually neutrons.

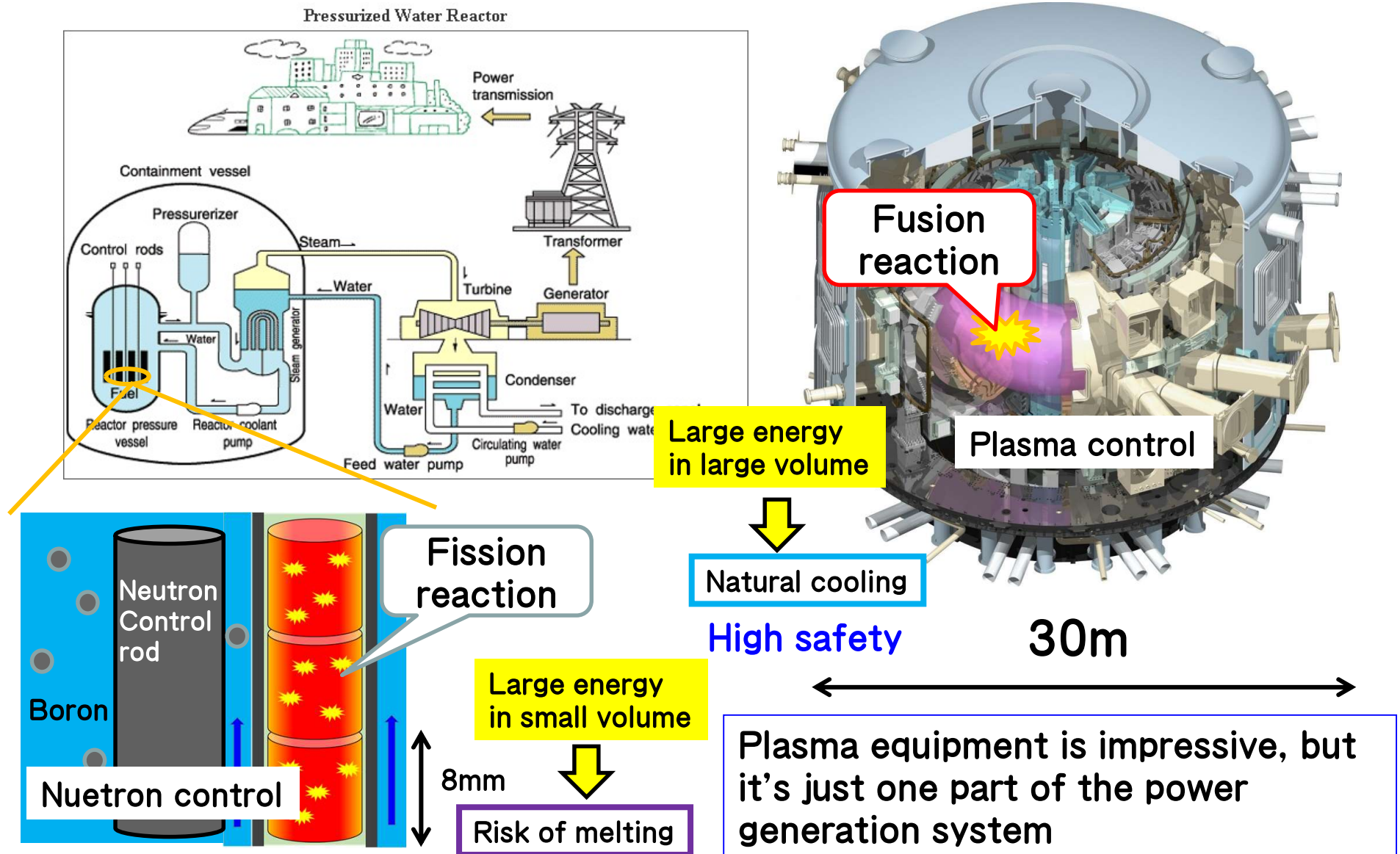


**Nuclear Fusion reaction:** The process of generating heat by joining two lightweight atoms to form a heavier atom.

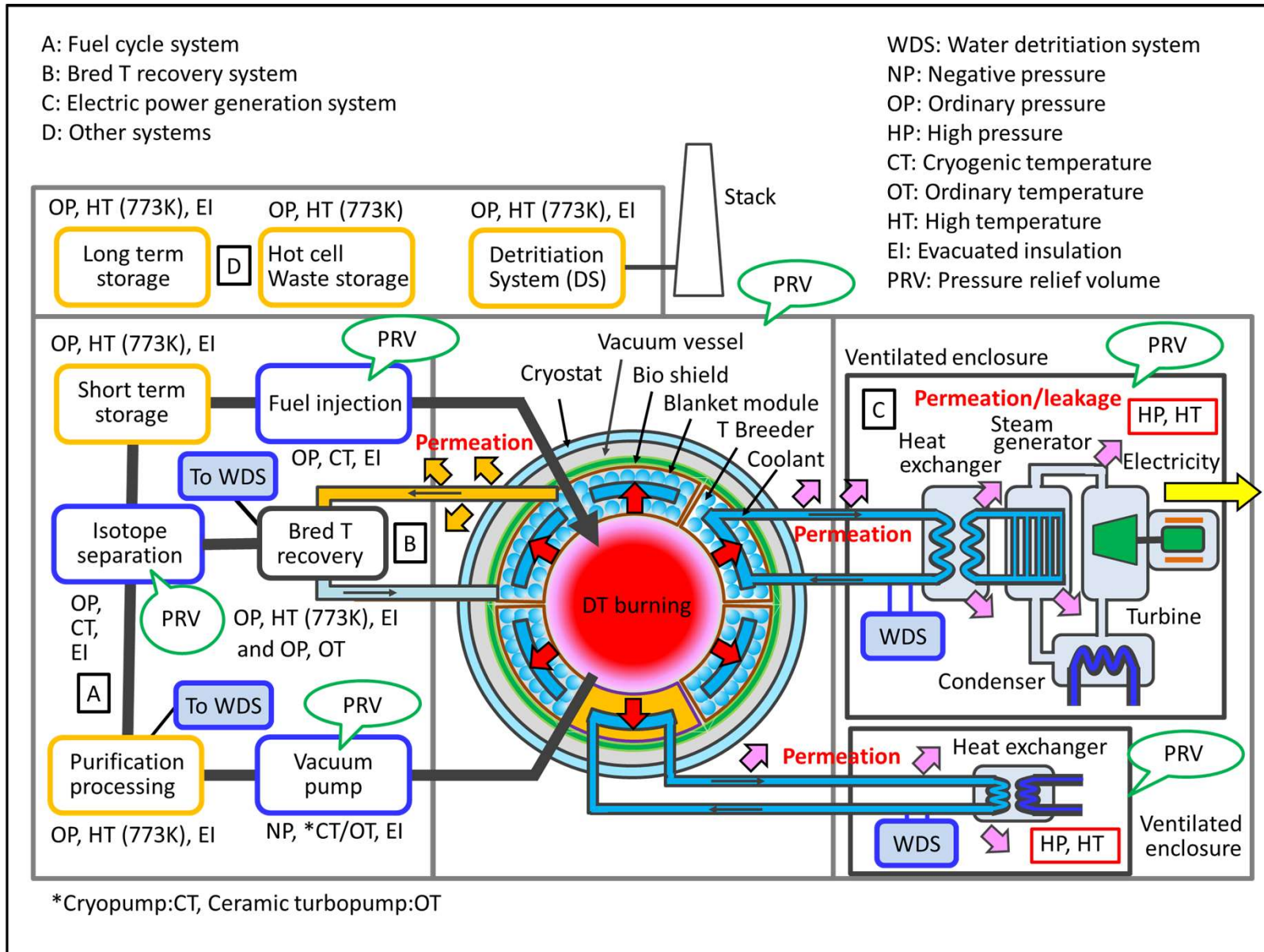
Energy from nuclear reaction (MeV) is million times larger than chemical reaction (eV) in thermal power generation.



# Reactor core in fission and fusion reactor

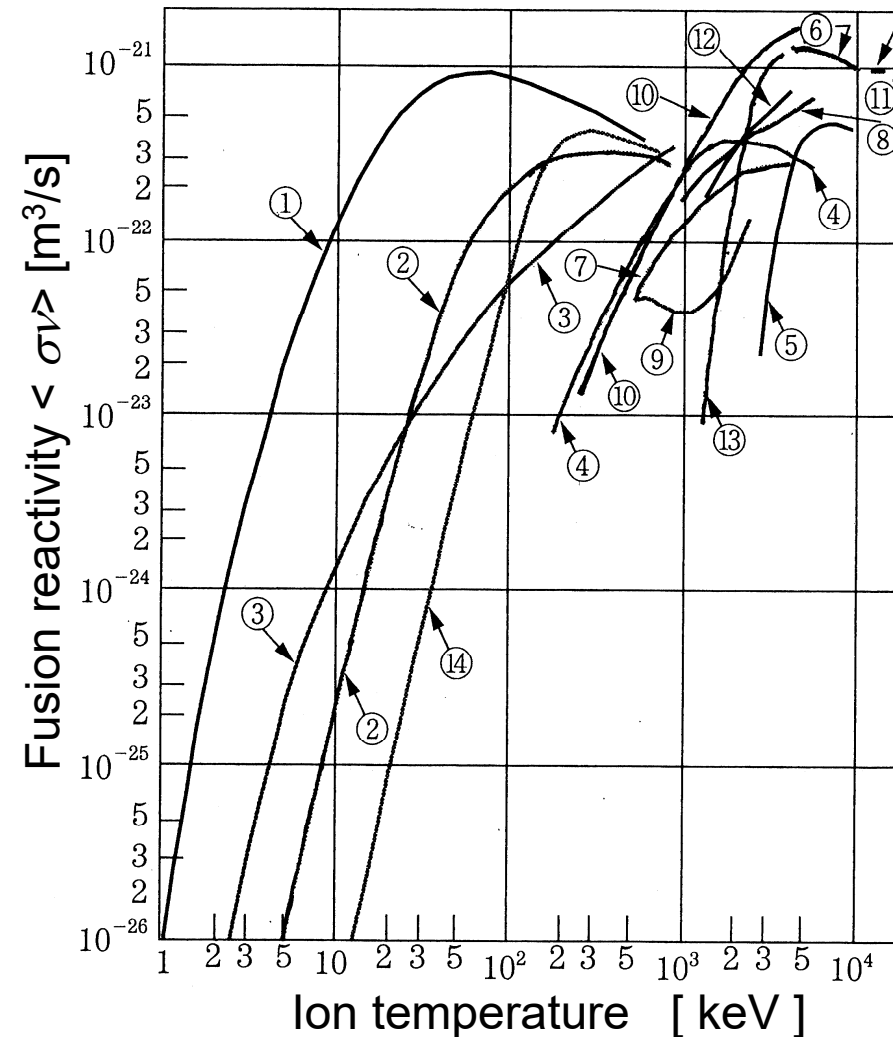
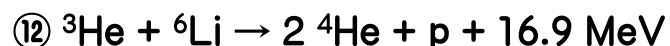
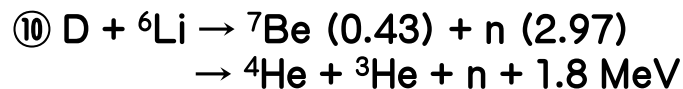
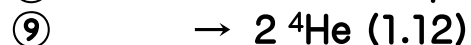
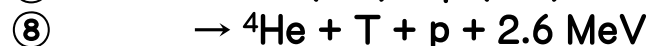
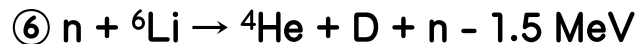
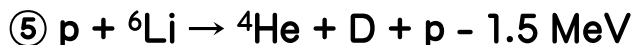
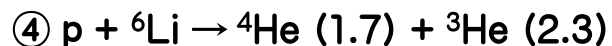
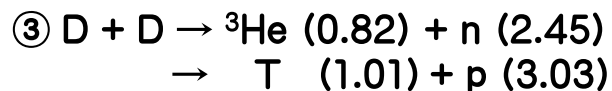
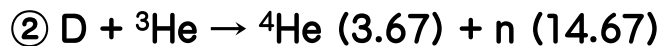


# Power generation system in fusion reactor



# Fusion reactions

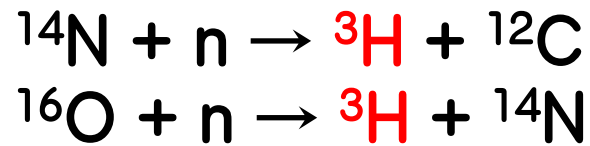
There are quite many different fusion reactions. D-T reaction requires the least energy to give the highest cross section at around a few 10's keV. Therefore, realization of a D-T fusion reactor is the principal goal.



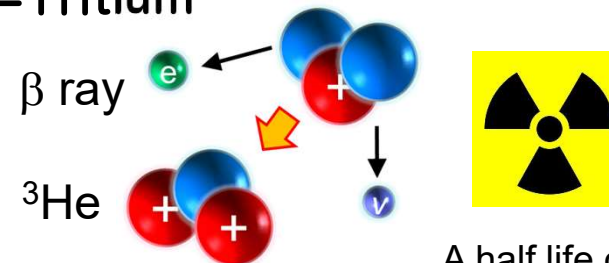
# Tritium in nature

Deuterium is present in seawater at 150 ppm, but Tritium is little in nature.

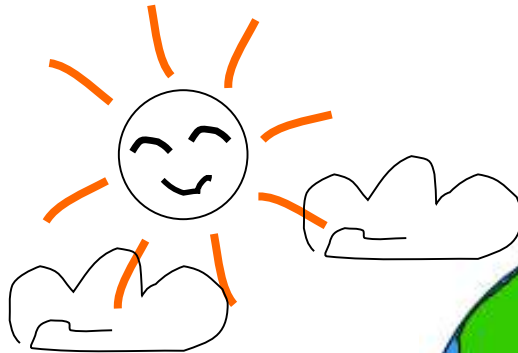
T is naturally produced by the following reaction with cosmic rays in the upper atmosphere.



${}^3\text{H}$ =Tritium

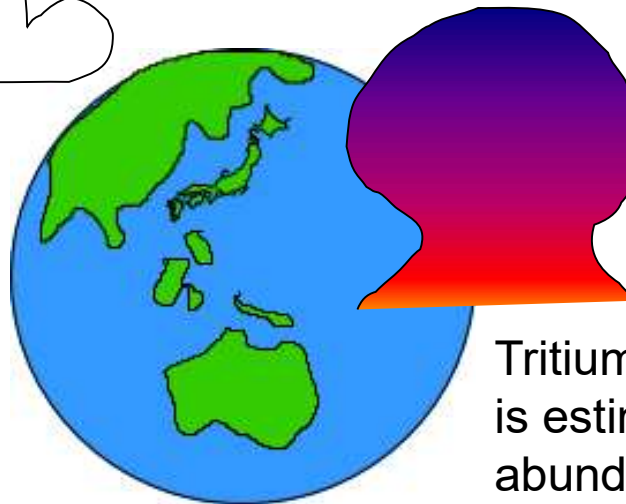


A half life of tritium  
is 12.3 years.



3000g of T in nature

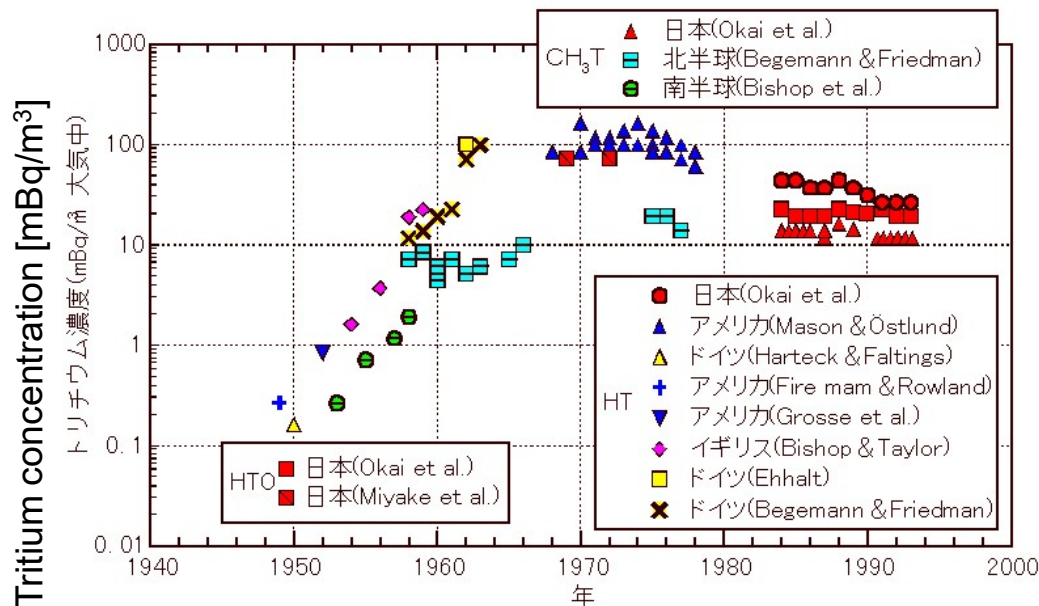
Atmospheric nuclear weapon test  
(1960s)



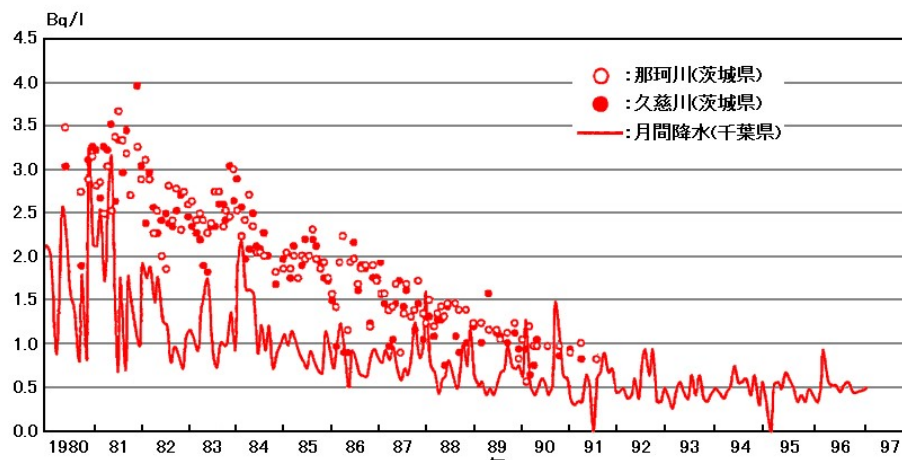
Tritium released in nuclear tests  
is estimated to be 200 times the  
abundance in nature.



# Tritium concentration in nature



Change of atmospheric T concentration with time



Changes of tritium concentrations in rainwater and river water with time (Japan)

By nuclear weapon test, tritium concentration in nature was increased apparently in 1960s.

After nuclear weapon test ban, gradually tritium concentration decreased.

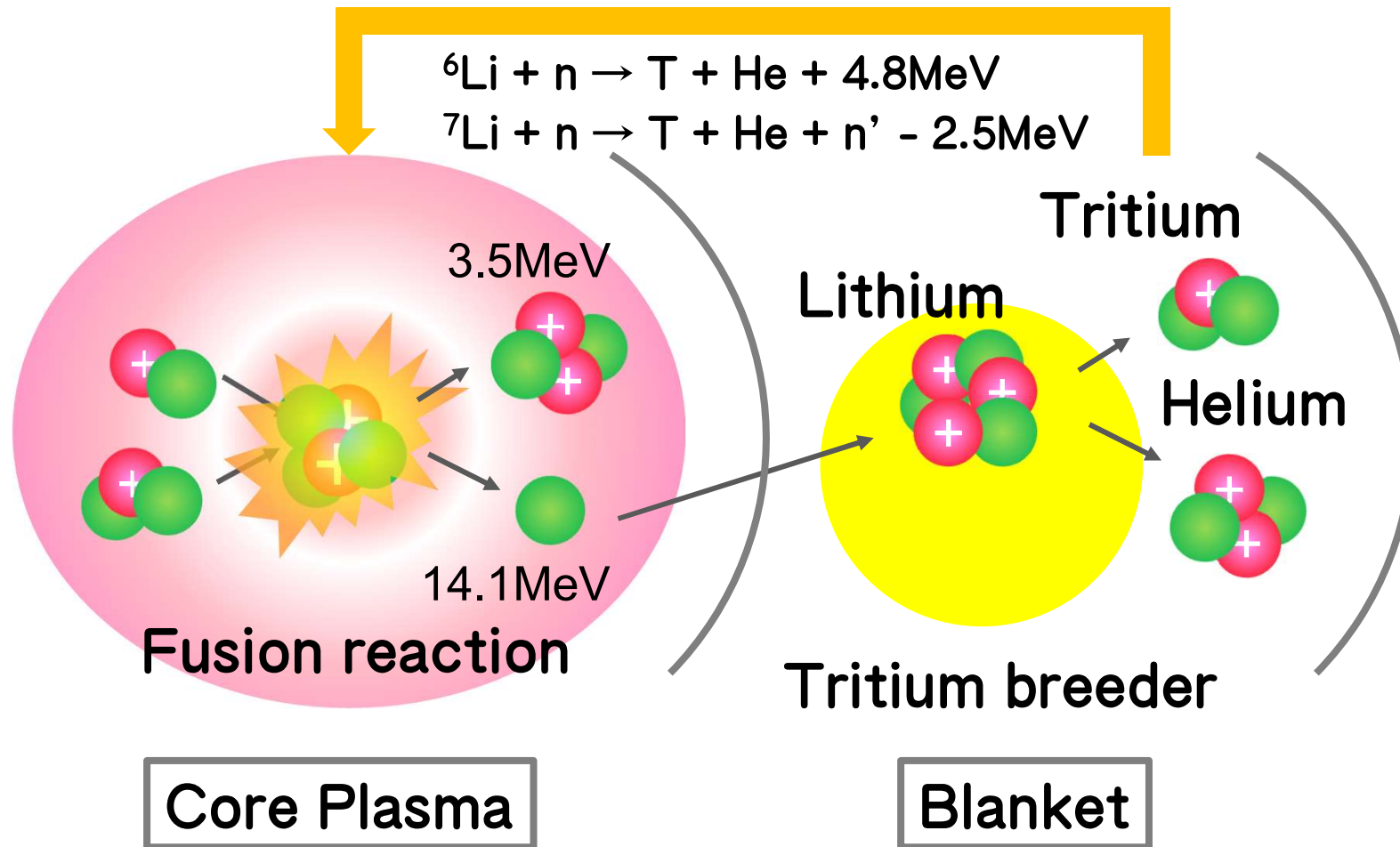
It is difficult to collect tritium from nature because its concentration is very low.

How do we produce tritium as fuel for a fusion reactor?



Nuclear reaction of neutron and Li is utilized.

# Tritium Production from Lithium



DT fusion  
T is lost

Neutron  
generation

n-Li reaction  
n is lost

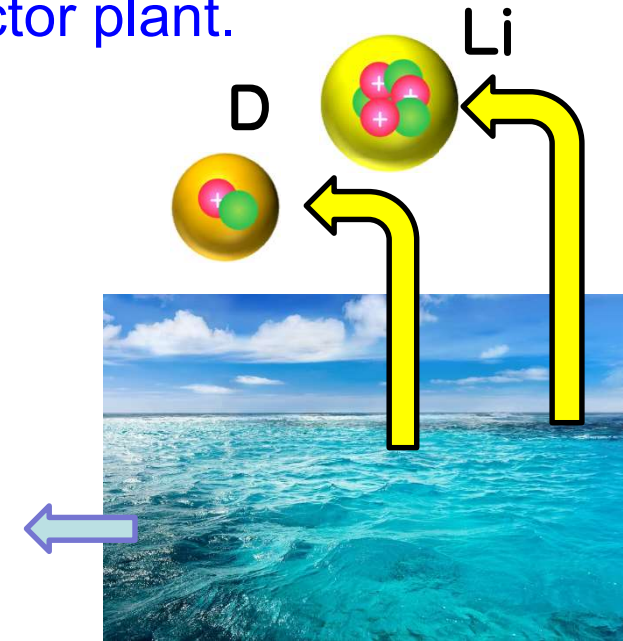
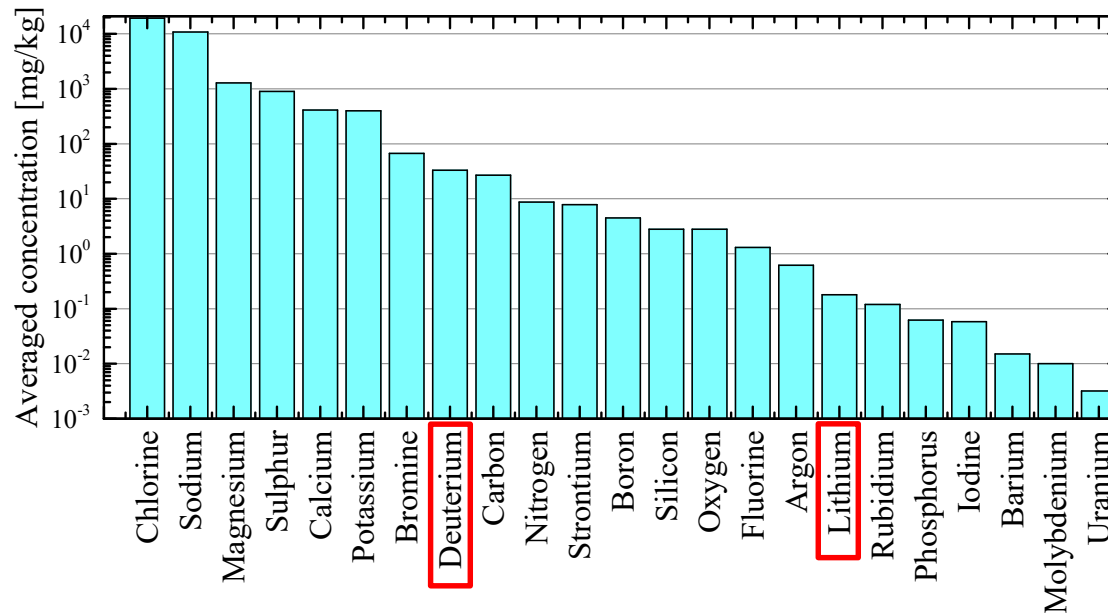
**T is  
generated**

# Fuel resource in seawater

It is generally agreed and believed that any fusion reactors based on the DT fuel cycle will need to be self-sufficient in tritium fuel because **tritium cannot be acquired from natural sources**.

Fuel tritium is self-produced in the fusion reactor plant.

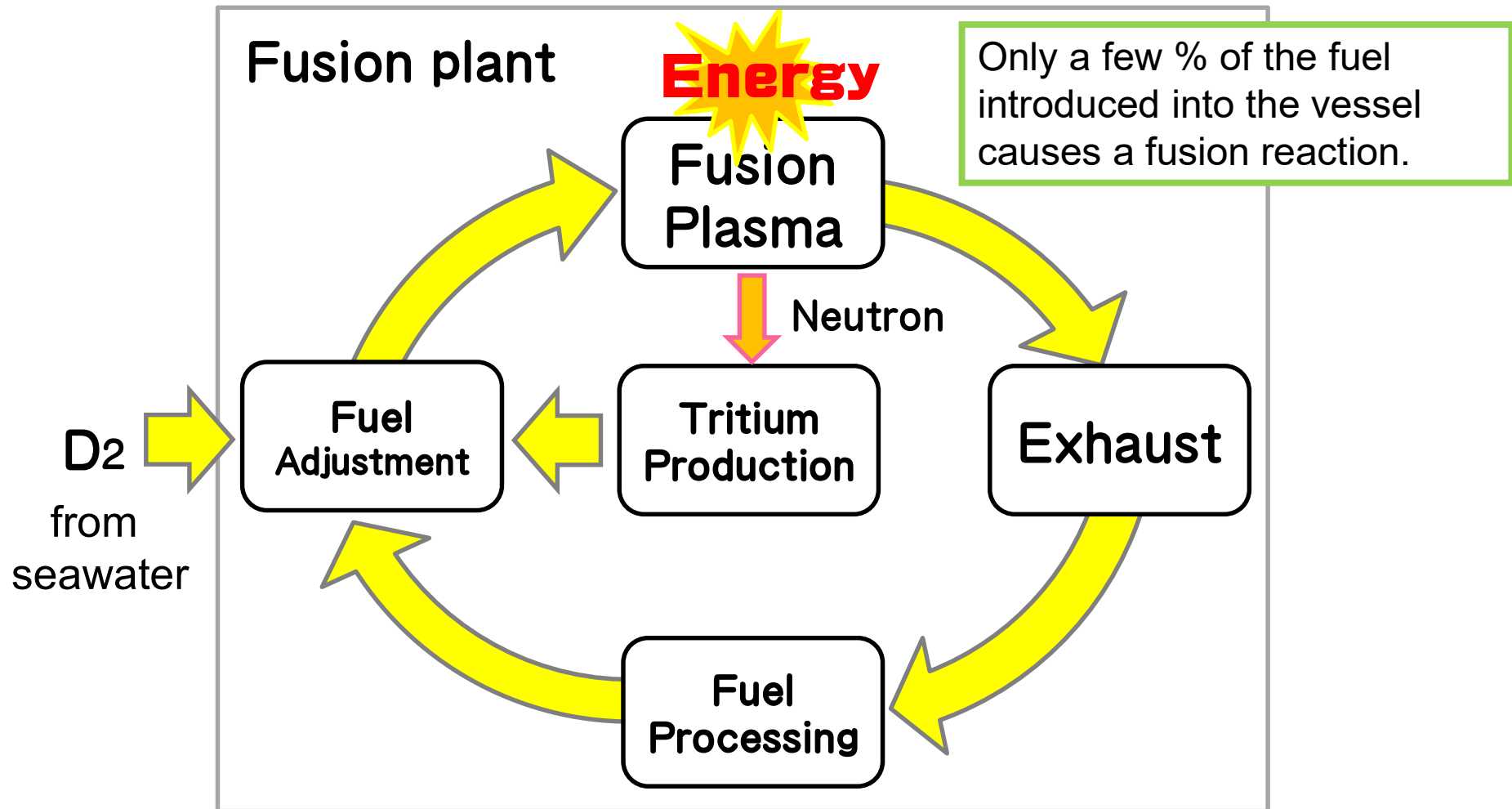
Energy resource of fusion reactors is deuterium and lithium.



Concentrations of D and Li is relatively high.

D<sub>2</sub>:150ppm  
Li :0.5ppm

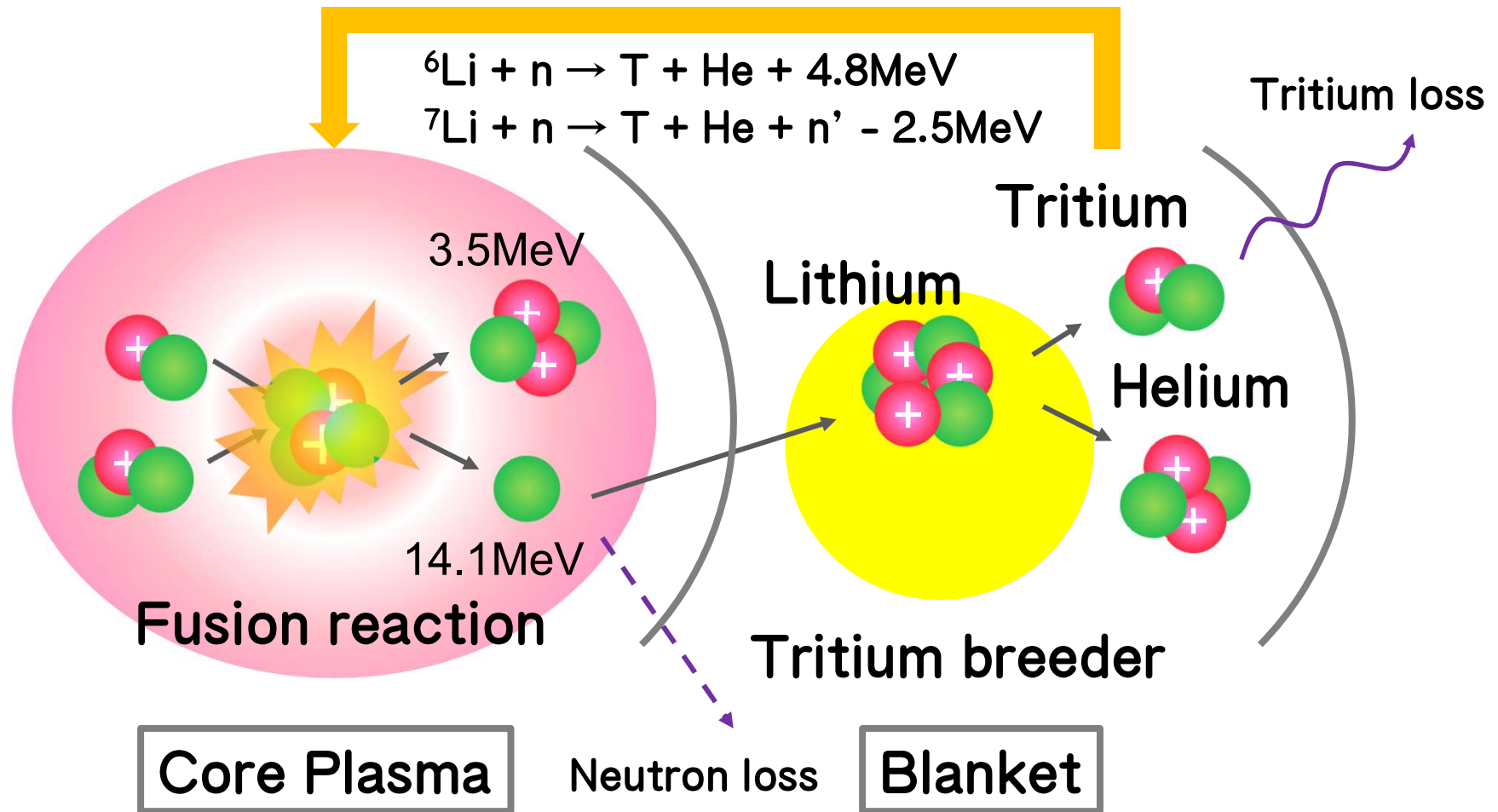
# Fuel Cycle & Tritium Production



Tritium cannot be acquired from natural sources. Therefore, tritium is produced continuously in the fusion plant.



# Tritium Production from Lithium



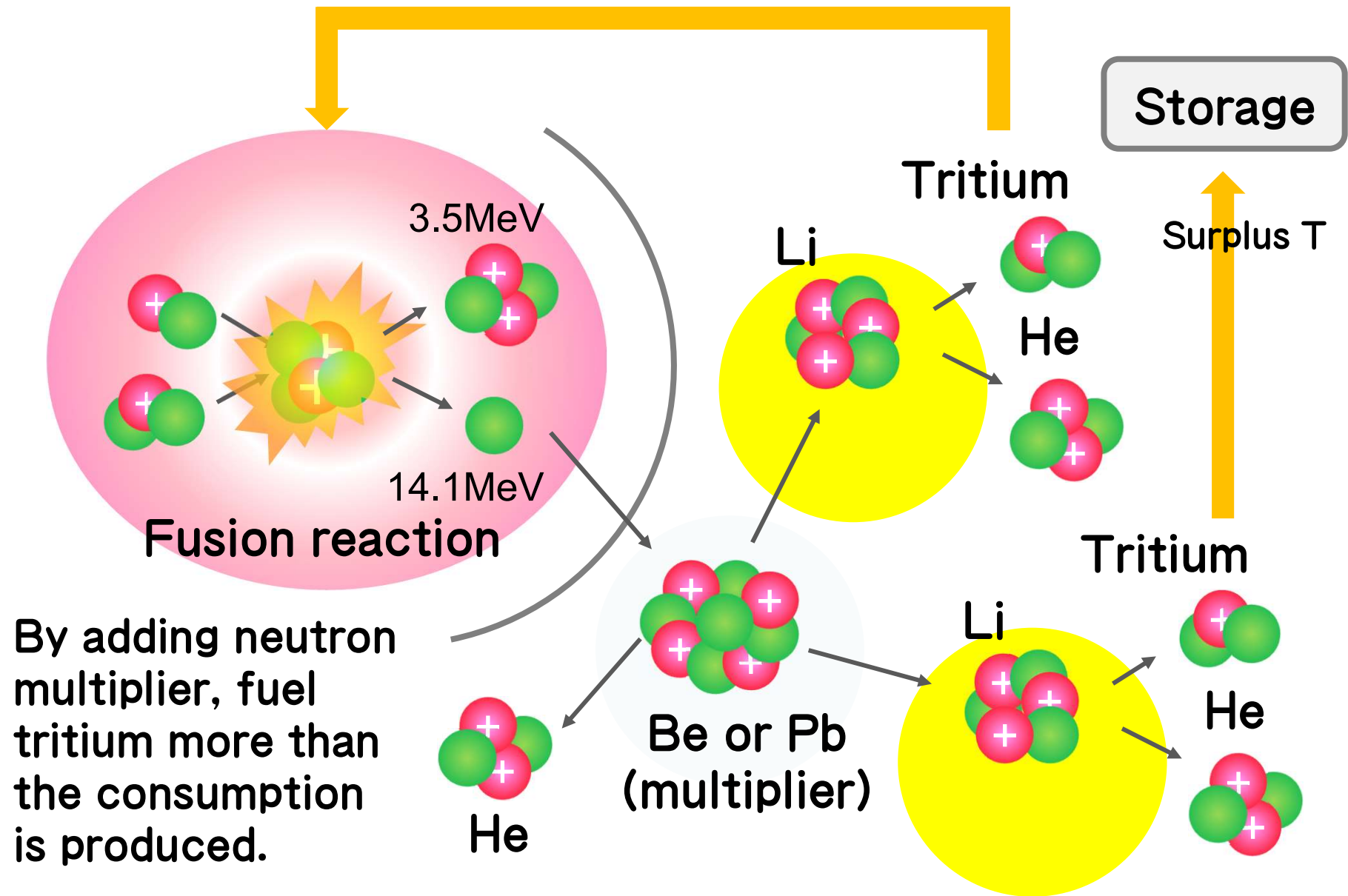
DT fusion  
T is lost

Neutron  
generation

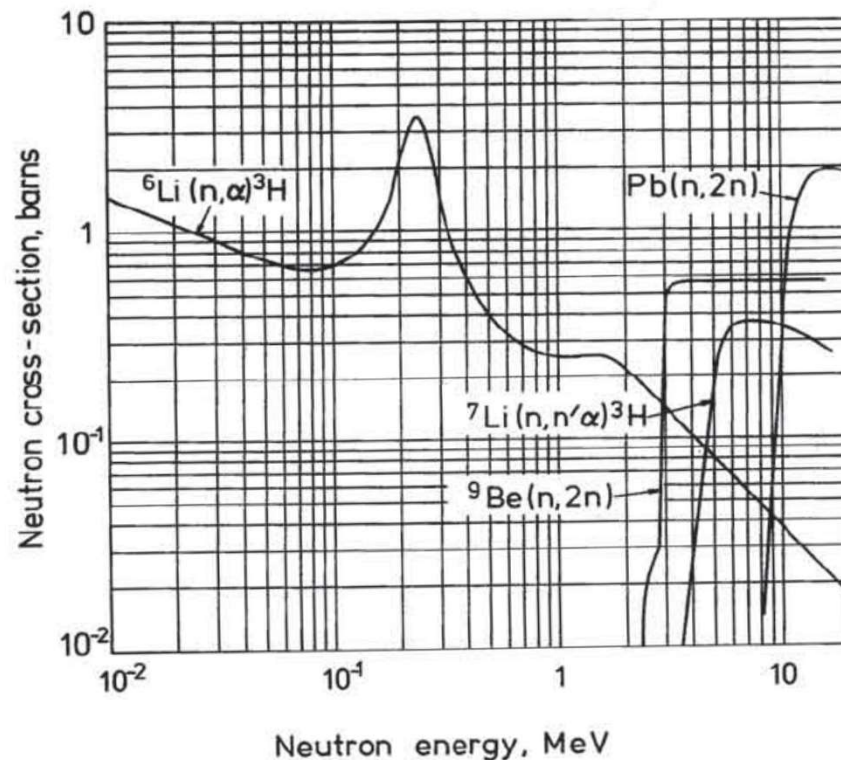
n-Li reaction  
n is lost

**T is  
generated**

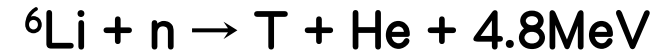
# Neutron Multiplier



# Neutron reaction cross-section

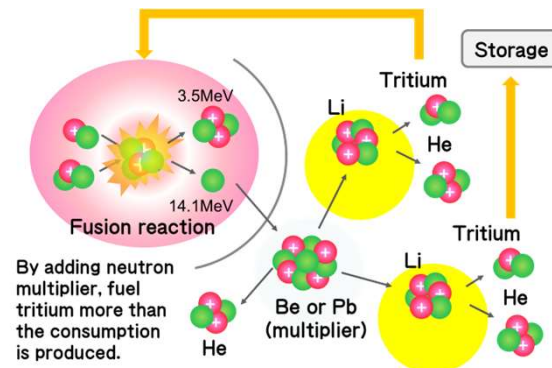


## Tritium production reactions



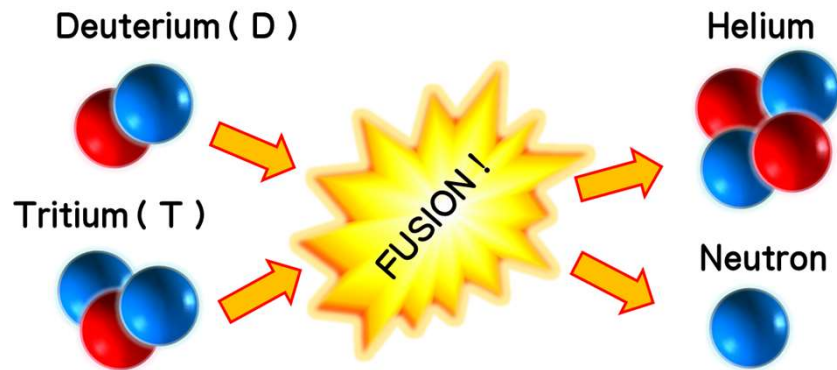
In nature,  ${}^6\text{Li}$  and  ${}^7\text{Li}$  are present in a ratio of 7.5 : 92.5.

## (n, 2n) reactions



High energy neutron reacts with Beryllium or Lead and low energy neutrons are created. The low energy neutrons mainly reacts with  ${}^6\text{Li}$ . Isotopic enrichment is required.

# D-T fusion reaction



Total energy production  
17.6 MeV

## Question!

How many grams of **deuterium** and **tritium**, are consumed per day in a 3GW thermal power fusion plant?

(3 GW thermal power is equivalent to 100 MW electric power.)

- 17.6 MeV energy is generated by a fusion reaction.
- Tritium 1mol = 3 g, Deuterium 1mol = 2 g
- 1 eV is  $1.6 \times 10^{-19}$  J
- 1 mol is  $6.0 \times 10^{23}$  atoms



# Tritium and Deuterium consumption

How many grams of **Deuterium** and **Tritium**, are consumed per day in a 3GW thermal power fusion plant?

- 17 MeV energy is generated by a fusion reaction.
- Tritium 1mol = 3 g, Deuterium 1mol = 2 g
- 1 eV is  $1.6 \times 10^{-19}$  J
- 1 mol is  $6.0 \times 10^{23}$  atoms

## Deuterium

$$\frac{3 \times 10^9 \text{ J/s} \times 3600 \text{ s/h} \times 24 \text{ h}}{17.6 \times 10^6 \text{ eV/fusion} \times 1.6 \times 10^{-19} \text{ J/eV}} \times \frac{2 \text{ g/mol}}{6 \times 10^{23} \text{ D/mol}} = 307 \text{ g/day}$$

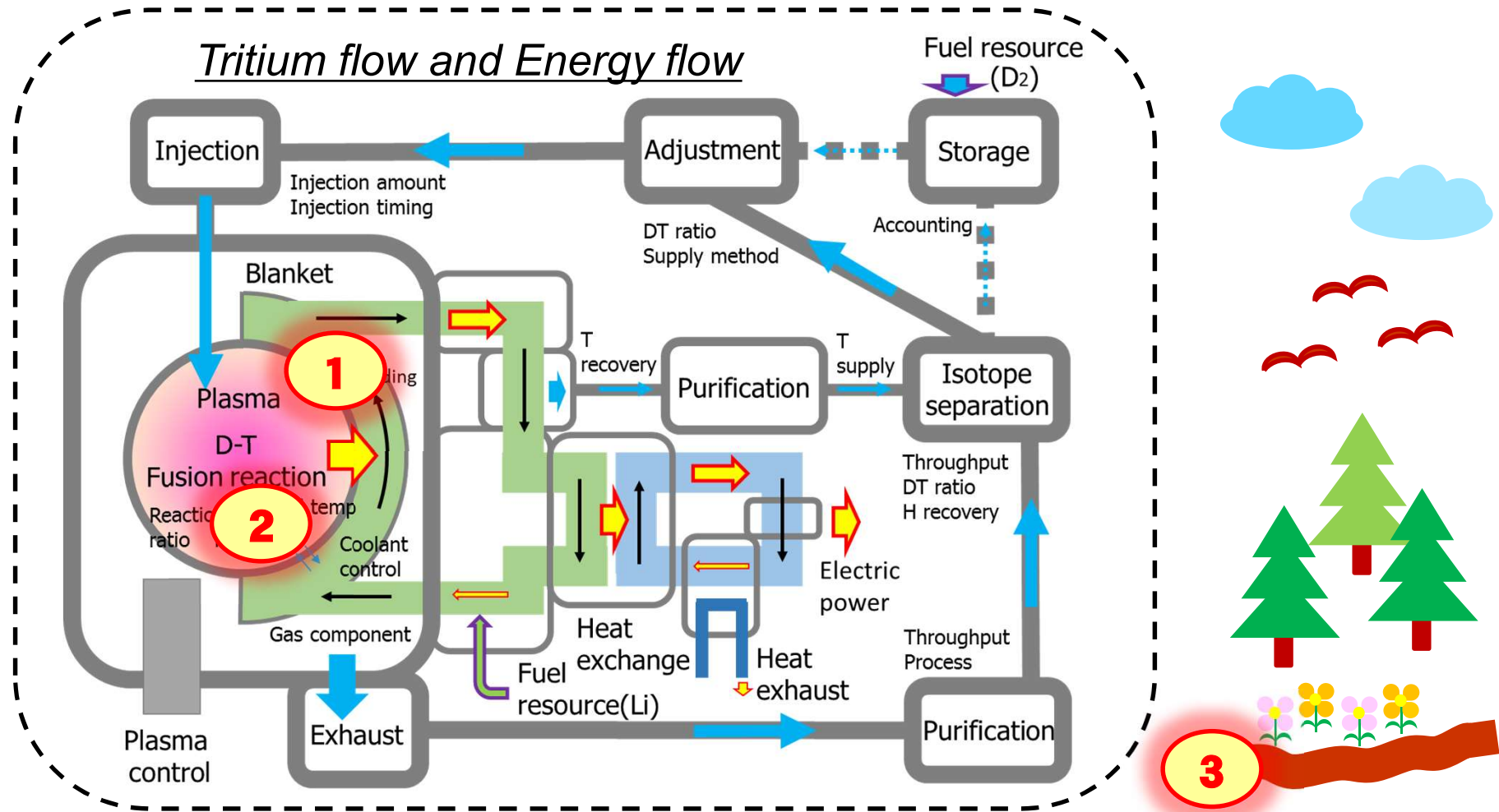
## Tritium

$$\frac{3 \times 10^9 \text{ J/s} \times 3600 \text{ s/h} \times 24 \text{ h}}{17.6 \times 10^6 \text{ eV/fusion} \times 1.6 \times 10^{-19} \text{ J/eV}} \times \frac{3 \text{ g/mol}}{6 \times 10^{23} \text{ T/mol}} = 460 \text{ g/day}$$

# Studies on Tritium mass transfer

1. Tritium production and extraction
2. Tritium retention and permeation

3. Tritium behavior in environment



# Tritium production in blanket

In order to produce tritium, tritium breeding materials are packed in the boxes installed in the plasma confinement vessel.

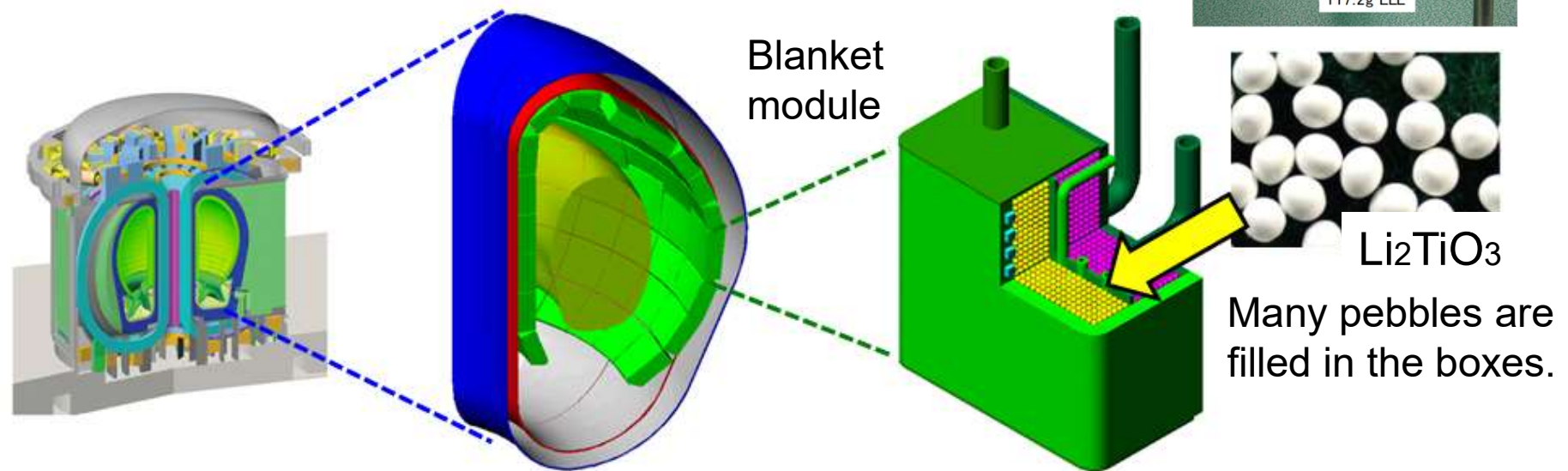
Understanding of release behavior of the bred tritium is important issues for the design of tritium extraction system and safety assessment.

LiPb

Tritium breeding materials;

**Solid breeder** :  $\text{Li}_2\text{TiO}_3$ ,  $\text{Li}_4\text{SiO}_4$ ,  $\text{Li}_2\text{O}$ ,  $\text{Li}_2\text{ZrO}_3$ ,  $\text{LiAlO}_2$

**Liquid breeder** : Li, LiPb, molten salt (Flibe, Flinabe)

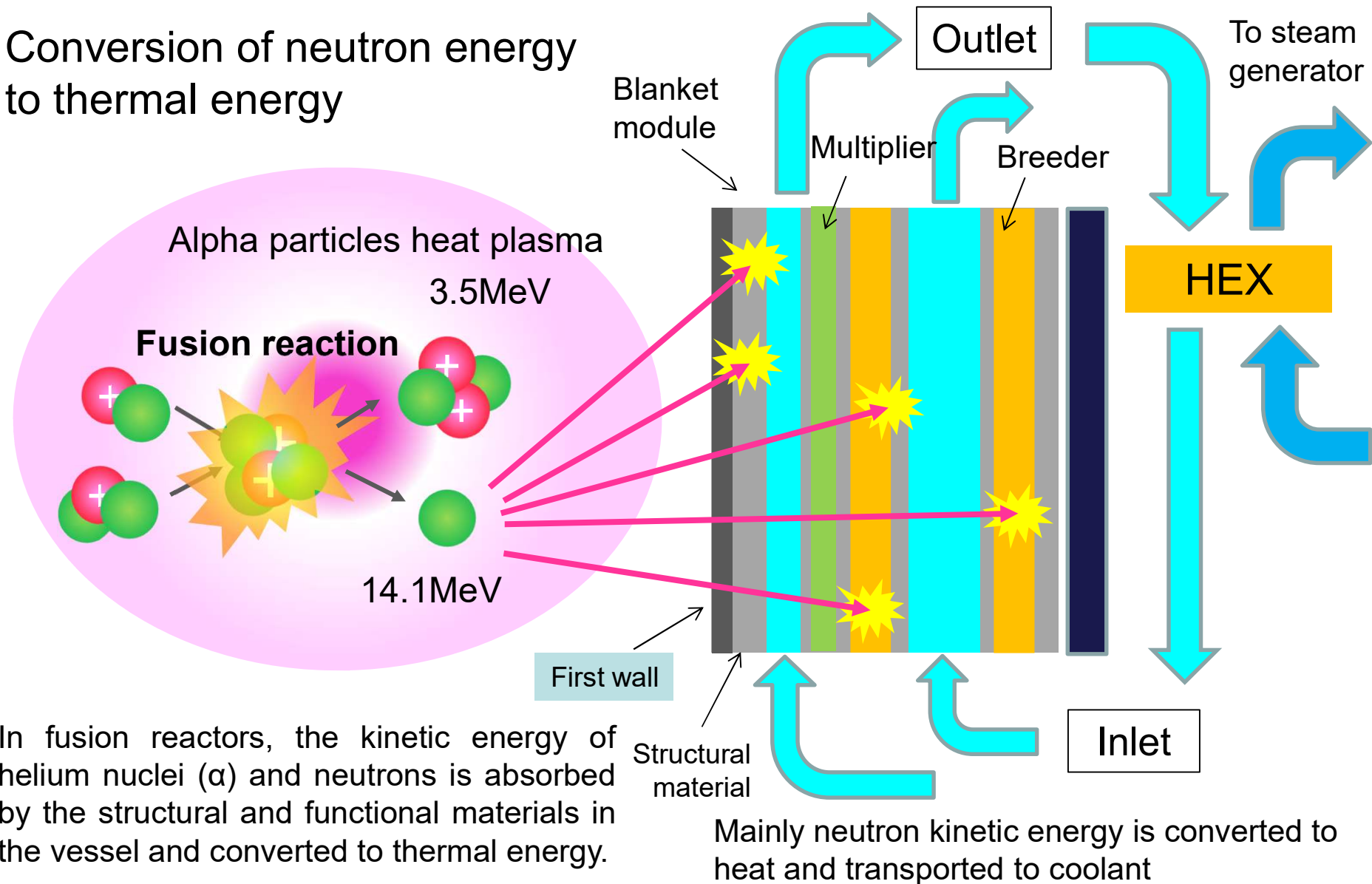


Illustrated by QST

<http://www.fusion.qst.go.jp/rokkasyo/en/project/blanket.html>

# Conversion of kinetic energy to heat

Conversion of neutron energy to thermal energy



In fusion reactors, the kinetic energy of helium nuclei ( $\alpha$ ) and neutrons is absorbed by the structural and functional materials in the vessel and converted to thermal energy.



# Neutron irradiation experiments

The candidate materials were irradiated by fission neutron in research reactor (JRR-4 or KUR).

Kyoto University Research Reactor (KUR)



$\text{Li}_2\text{TiO}_3$

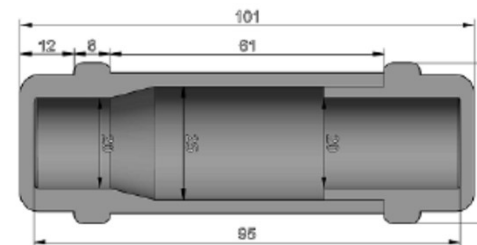


Li

FLiNaBe



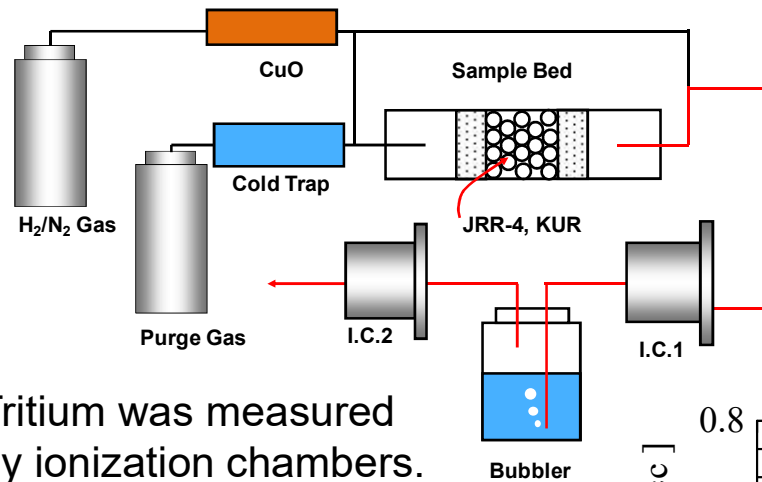
Neutron flux:  $5 \times 10^{12} \text{ n/cm}^2/\text{s}$   
Irradiation time: 5 min  
Irradiation temp:  $< 50 \text{ }^\circ\text{C}$



Irradiation capsule

# Tritium release from irradiated samples

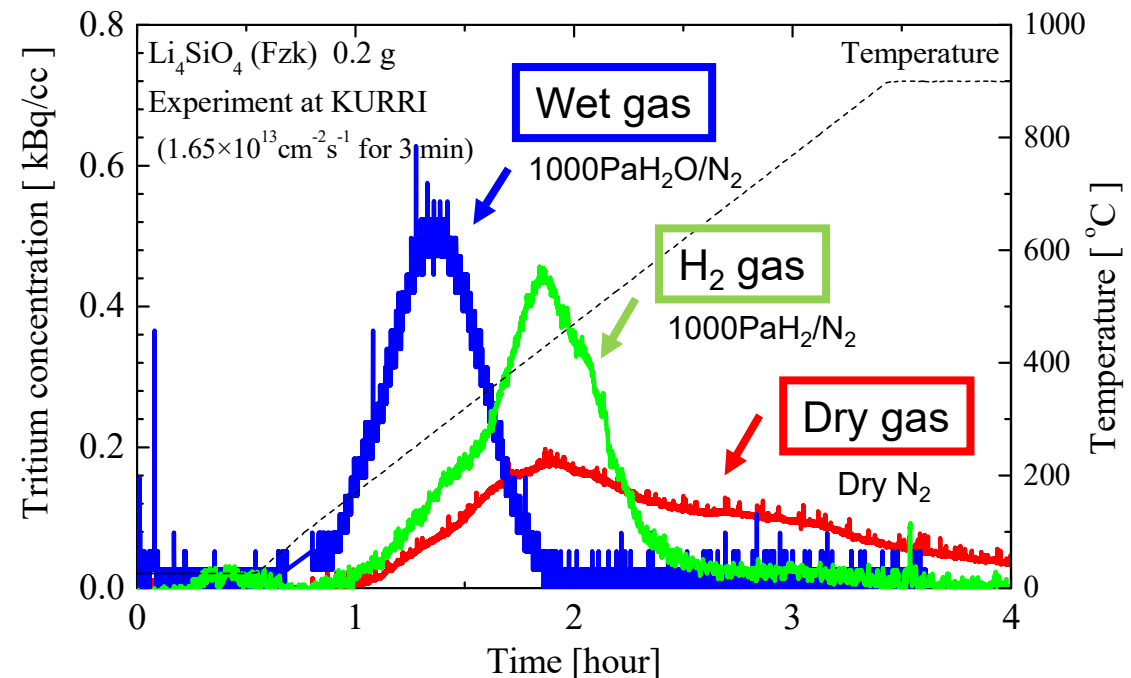
Tritium release from the pebbles of  $\text{Li}_2\text{TiO}_3$ .



Tritium was measured by ionization chambers.

Tritium release behavior is greatly different depending on the purge gas. This means surface reaction affects on tritium release behavior.

Neutron irradiated samples were packed in a quartz reaction tube and heated up to 900 °C under different purge gas conditions.



# Surface reaction of tritium on Breeder surface

The trapped tritium was released by dry or wet gas purge and monitored by Ionization chamber.

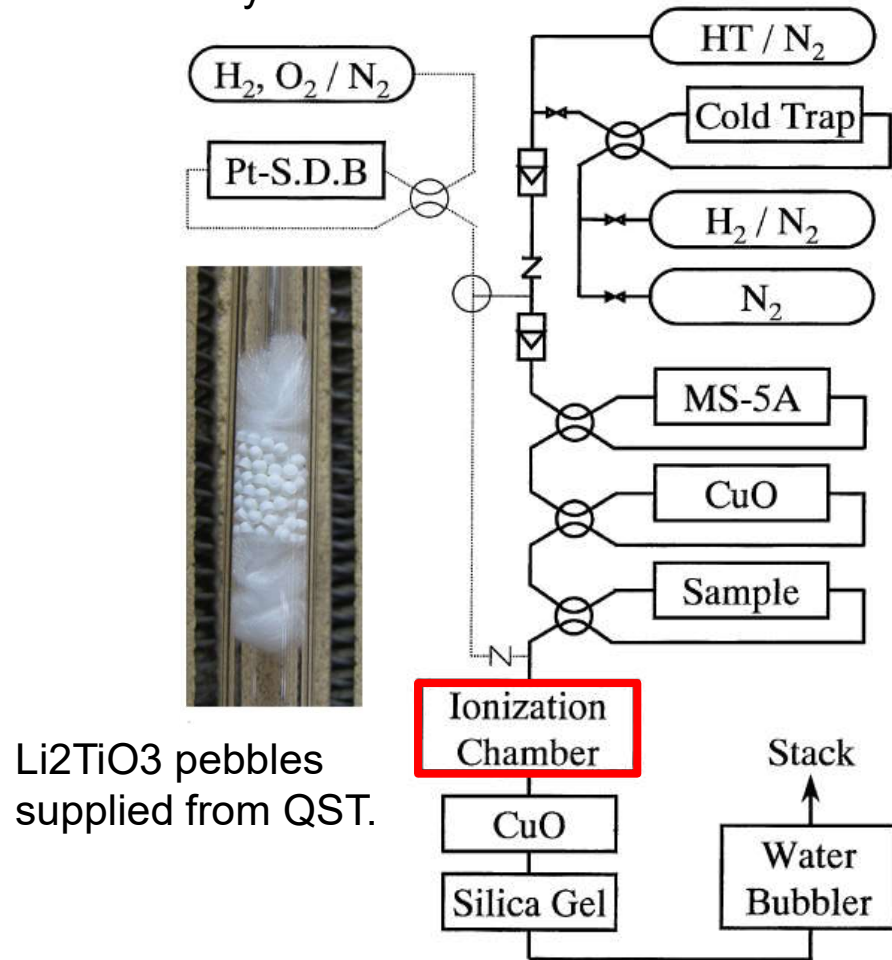
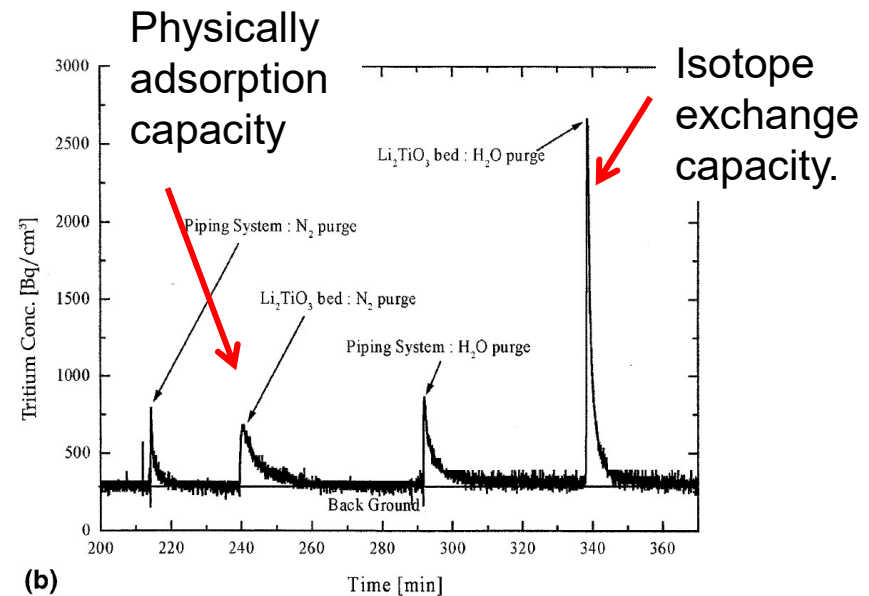
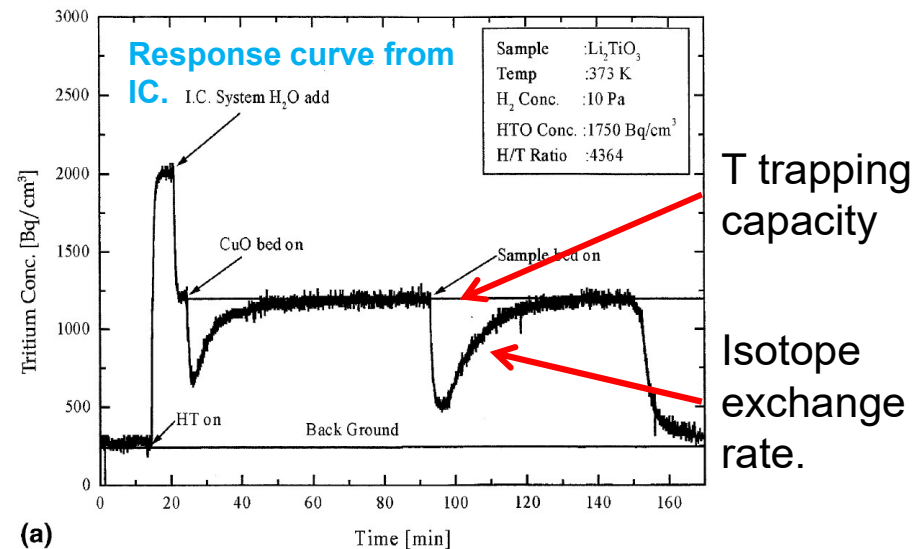


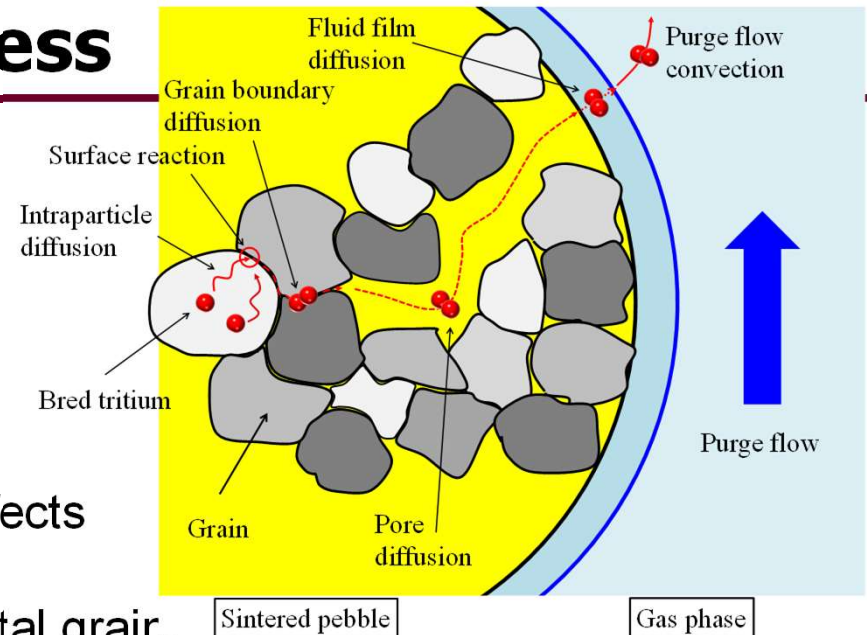
Fig. 1. Schematic diagram of the experimental apparatus.



# Tritium mass transfer process

T transfer processes and key reactions in overall T release property to a purge gas He containing H<sub>2</sub>.

1. T generation in a crystal grain.
2. **T diffusion** in the crystal grain.
3. Interaction of diffusing T with irradiation defects formed in the crystal grain.
4. **T transfer from inside to surface** of the crystal grain..
5. Absorption of T into the crystal grain.
6. **Adsorption/desorption** of HTO and H<sub>2</sub>O on crystal grain surface.
7. **Isotope exchange reaction** between H<sub>2</sub> in gas phase and T on the crystal grain surface (**isotope exchange reaction 1**).
8. **Isotope exchange reaction** between H<sub>2</sub>O in the gas phase and T on the crystal grain surface (**isotope exchange reaction 2**).
9. **Water formation reaction** on the crystal grain surface with H<sub>2</sub> in the gas phase.
10. Mass transfer of hydrogen isotopes and water vapor through the interconnected pores to geometrical surfaces of the pebbles.
11. Mass transfer of hydrogen isotopes and water vapor through fluid film formed between geometrical surfaces of the pebbles and the purge gas flow.

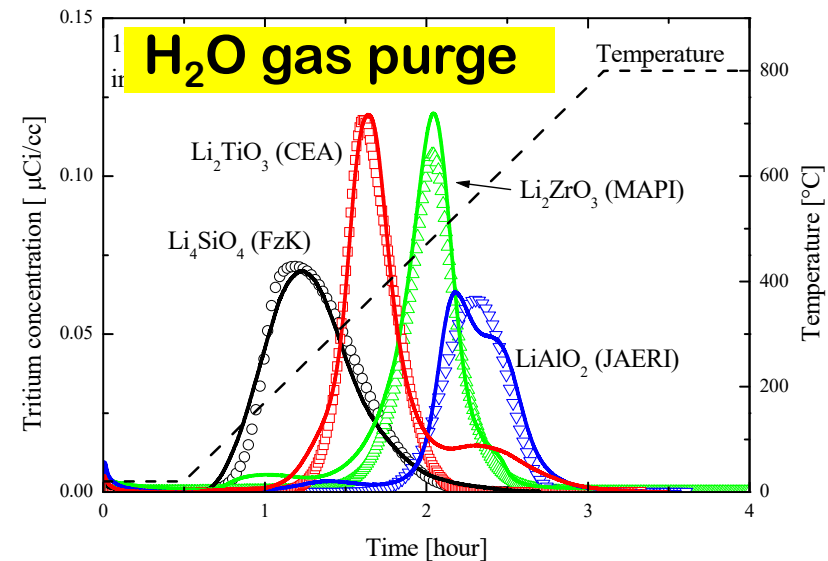
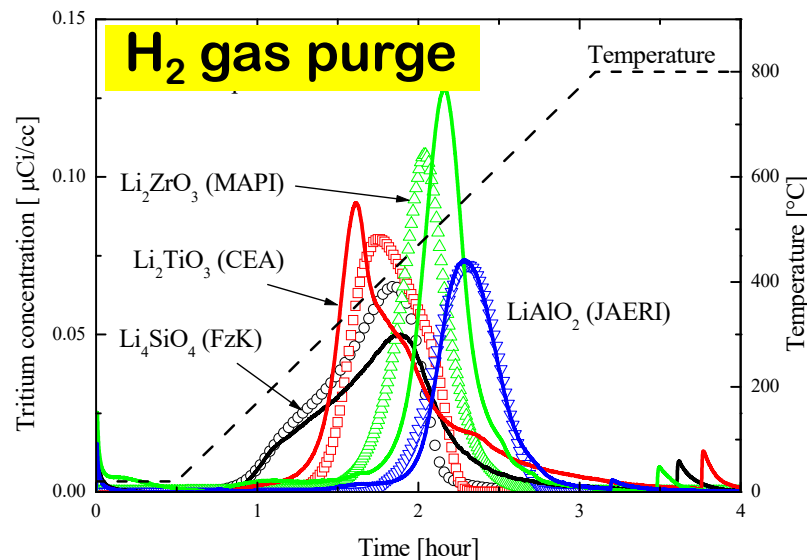
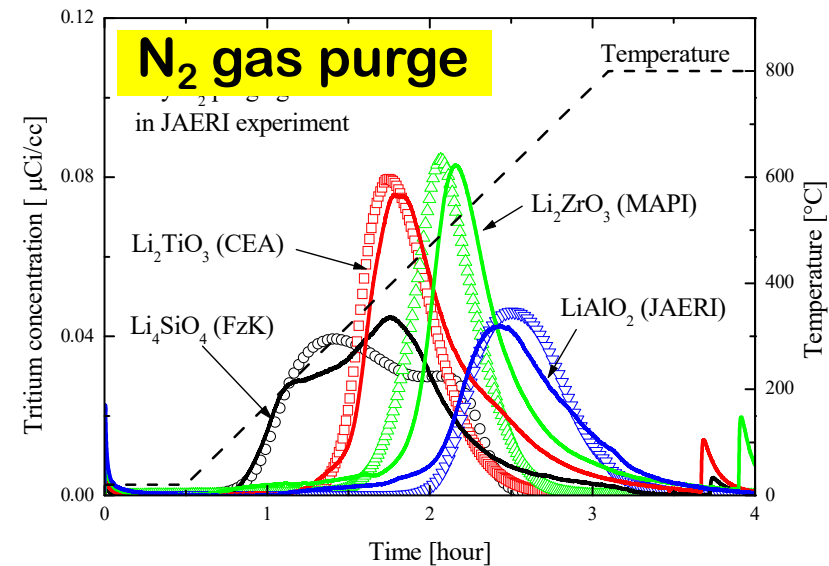




# Modeling and calculation

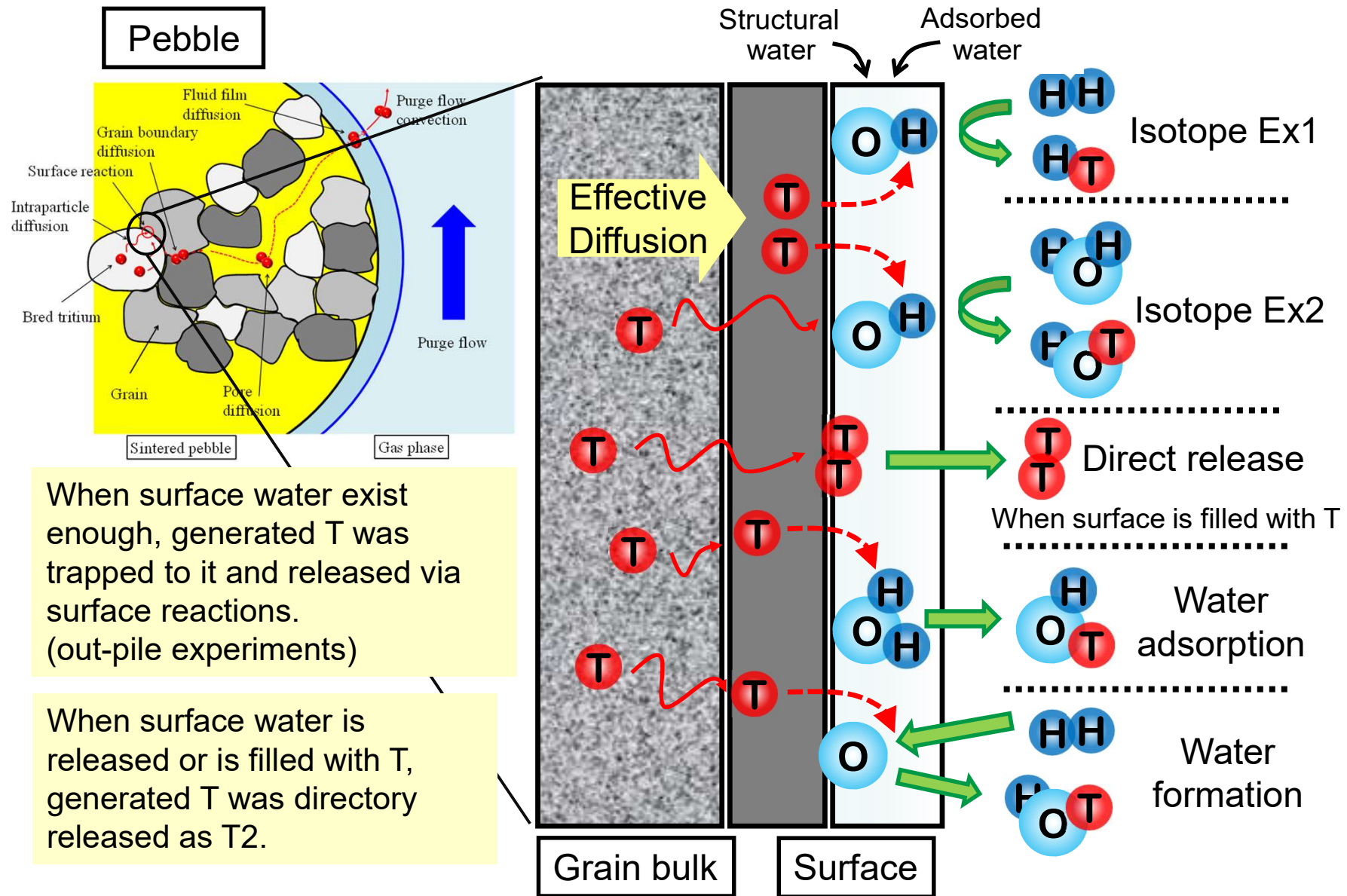
Numerical calculation based on the mass transfer model could reproduce experimental curves well.

Now, the research topic is proceeded to lithium evaporation at high temperature conditions.



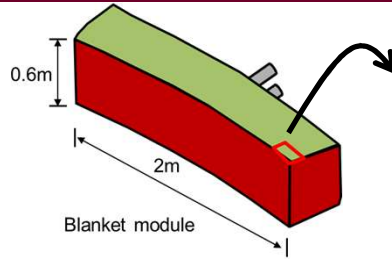


# Tritium release model in breeders



# Simulation of T behavior in the modules

# Simulation of tritium behavior in the blanket modules



Packed bed size: 200cm × 60cm × 5cm

Pipe: F82H, 160

Pipe size: OD1cm, Thickness:0.1cm

Center:900°C, Edge:300°C

$$(T=273+235+53185x-1.06e6x^2)$$

Ramping rate: 1°C/s

Purge gas: 100PaH<sub>2</sub>/He

Flow rate : 250slm (4.8cm/s)

## Pressure : Atmospheric

Coolant flow: 3m/s(No-dissolved H<sub>2</sub>)

Breeder: Li<sub>2</sub>TiO<sub>3</sub> (CEA, NFI)

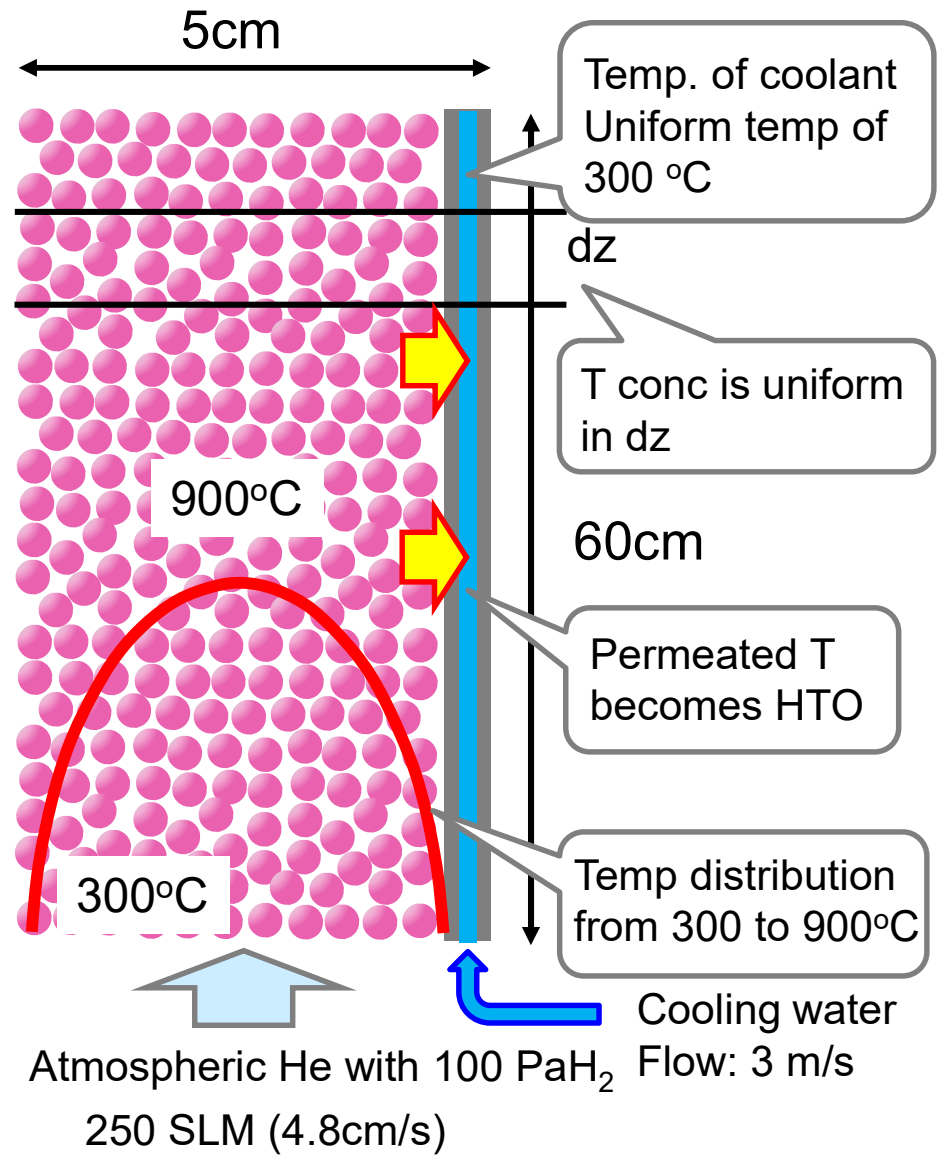
Grain size:  $1\mu\text{m}$

Pebble size: 1mm

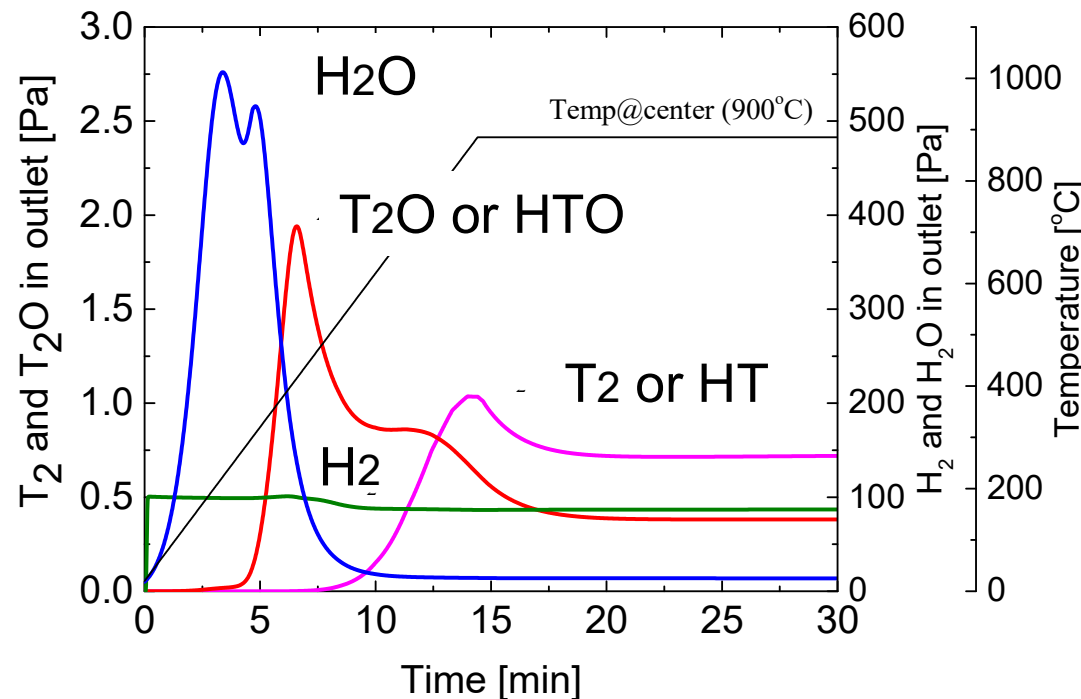
Density : 2.84g/cm<sup>3</sup>(83%TD)

Packing ratio: 65%

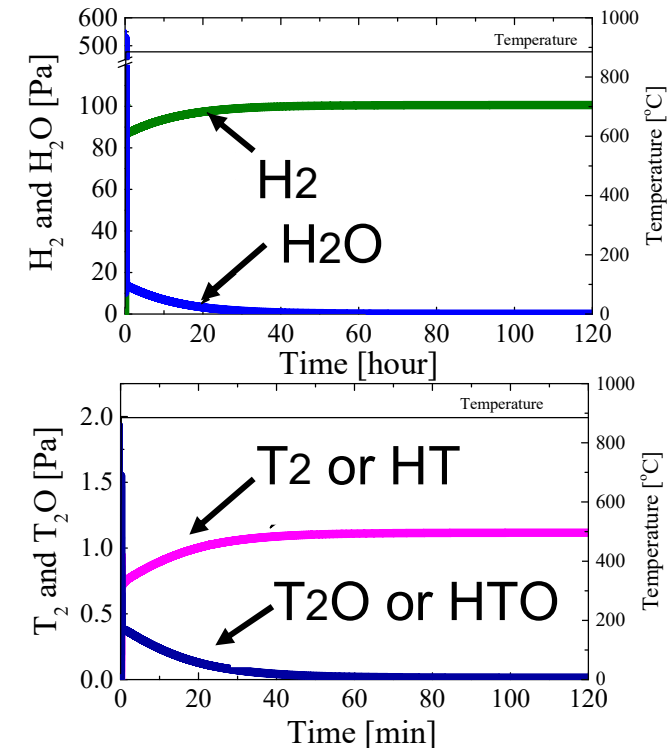
T generation rate:  $5 \times 10^{13}$  T/cm<sup>3</sup>/s



# Simulation of T behavior in the modules



First 30 minute

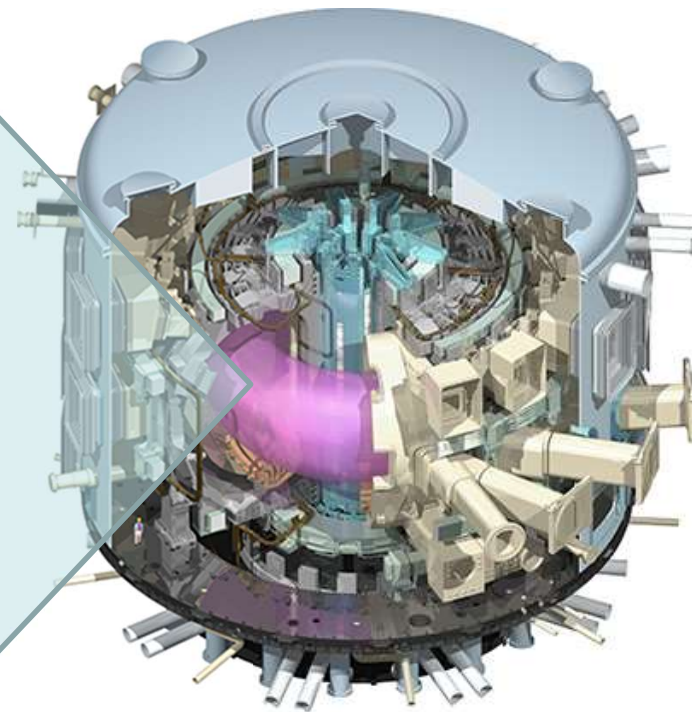
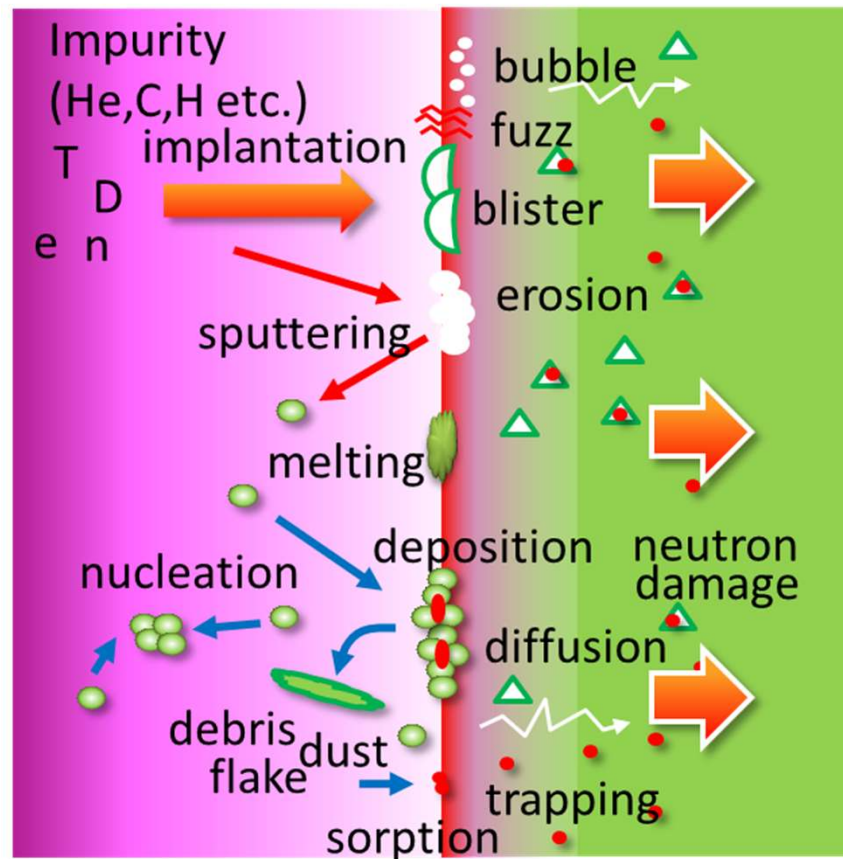


First 5 days

1. Large amount of water is released depending on temperature. ( $\sim 600^{\circ}\text{C}$ )
2. Diffused T reaches on the surface and it is released by Iex reaction 2. ( $T > 300^{\circ}\text{C}$ )
3. T on the surface is released by Iex reaction 1. ( $T > 500^{\circ}\text{C}$ )
4. Release of adsorbed water is completed but generated water is released.

In the longtime operation, T is released as T2 or HT

# Plasma – wall interaction

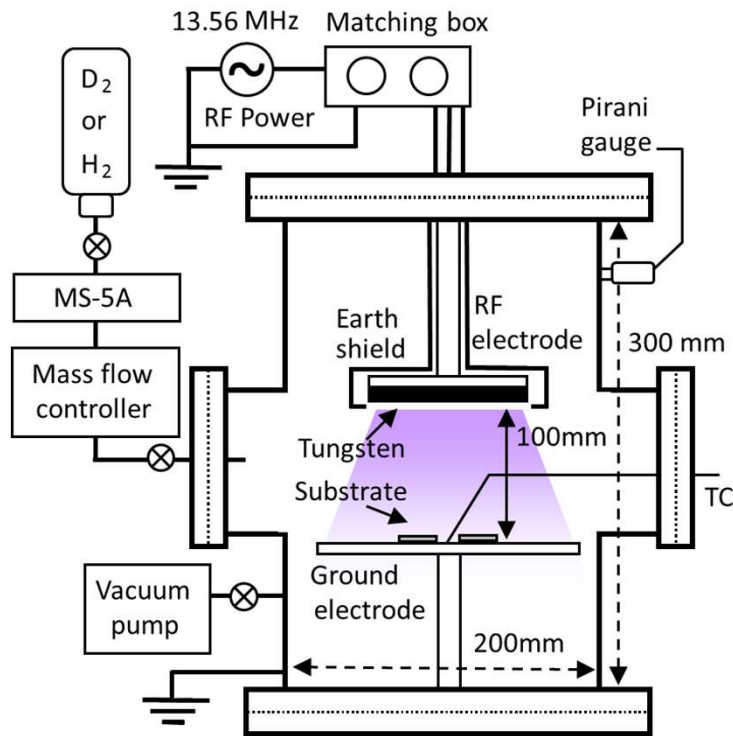


Carbon wall  $\Rightarrow$  Tungsten wall

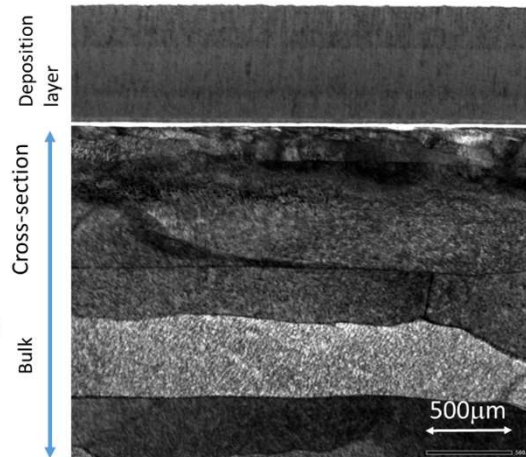
Understanding of tritium behavior in plasma facing materials is an important issue from viewpoints of fuel control and tritium safety. In this study, fuel accumulation into tungsten deposition layer was investigated by using lab experiments and large plasma devices.



# Hydrogen retention in W deposition layer



Capacitive Coupled Plasma

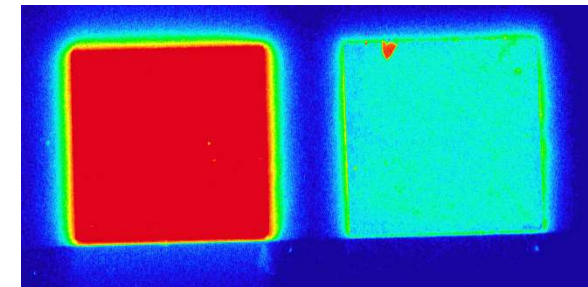


SEM observation

- High level of tritium was retained in only deposition layer.

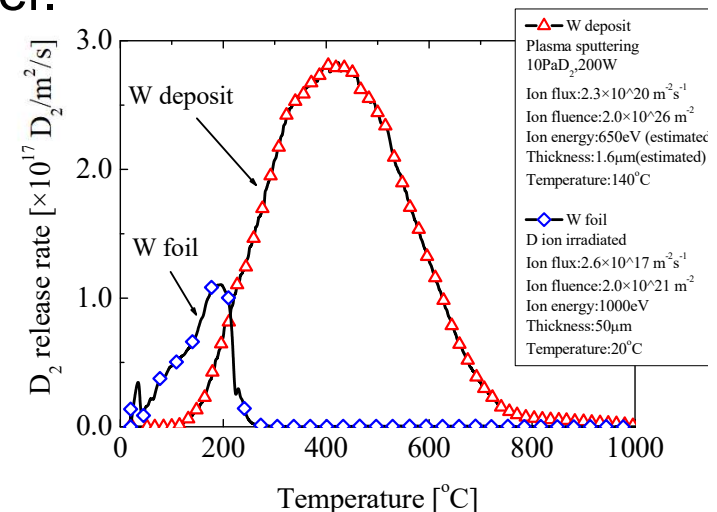
Tritium gas exposure Experiment.

W deposition W bulk



Deuterium release from W deposition and W bulk.

W deposition layer consists of fine grains with 2~3 nm. A large amount of hydrogen isotopes can be retained in grain boundaries.





# High energy plasma exposure

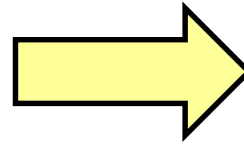


EAST in ASIPP (China)



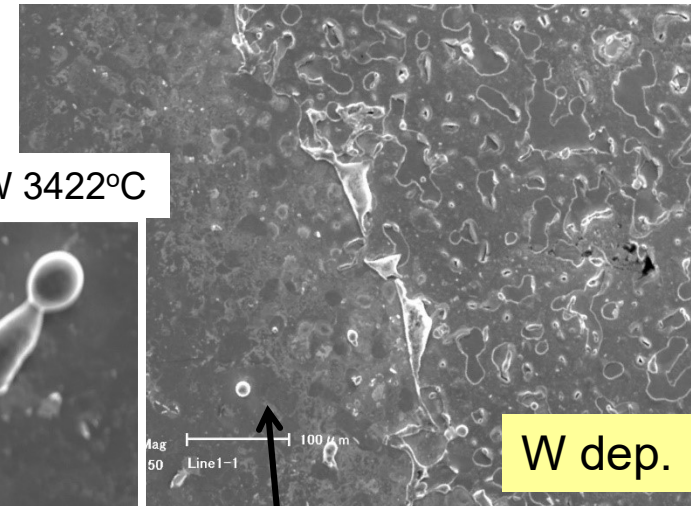
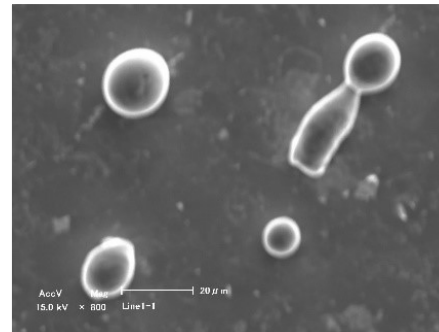
Many achievements were obtained  
international collaboration.

KSTAR  
in NFRI  
(Korea)



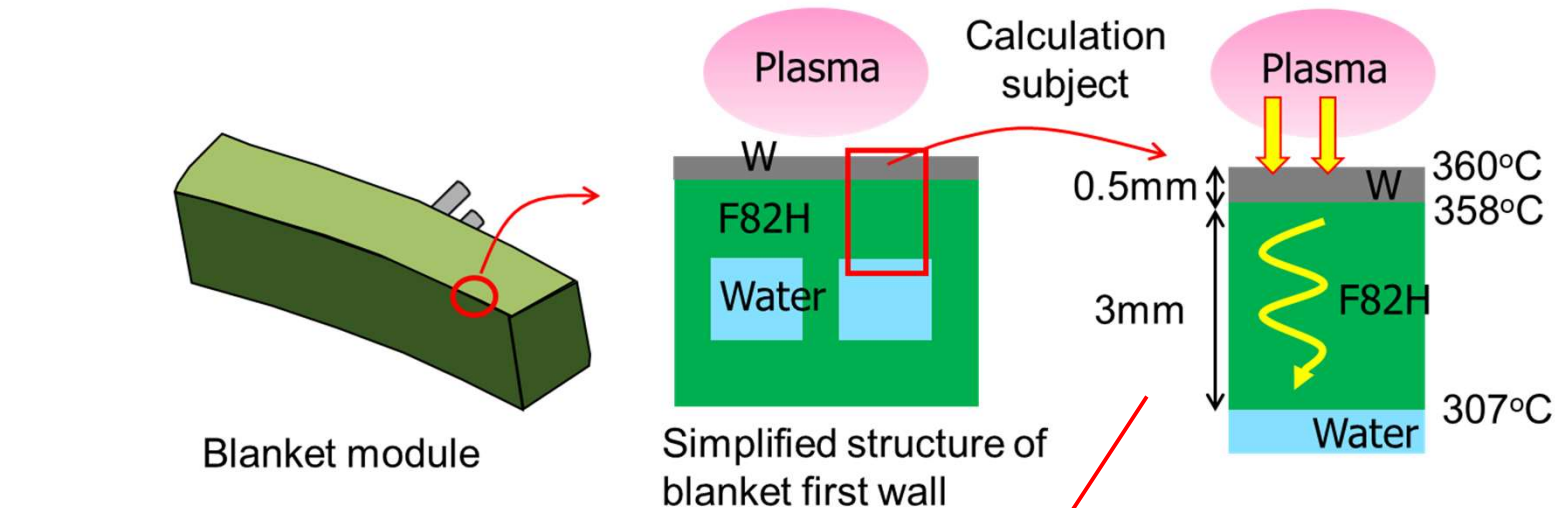
Melting·  
Evaporating

Melting point of W 3422°C

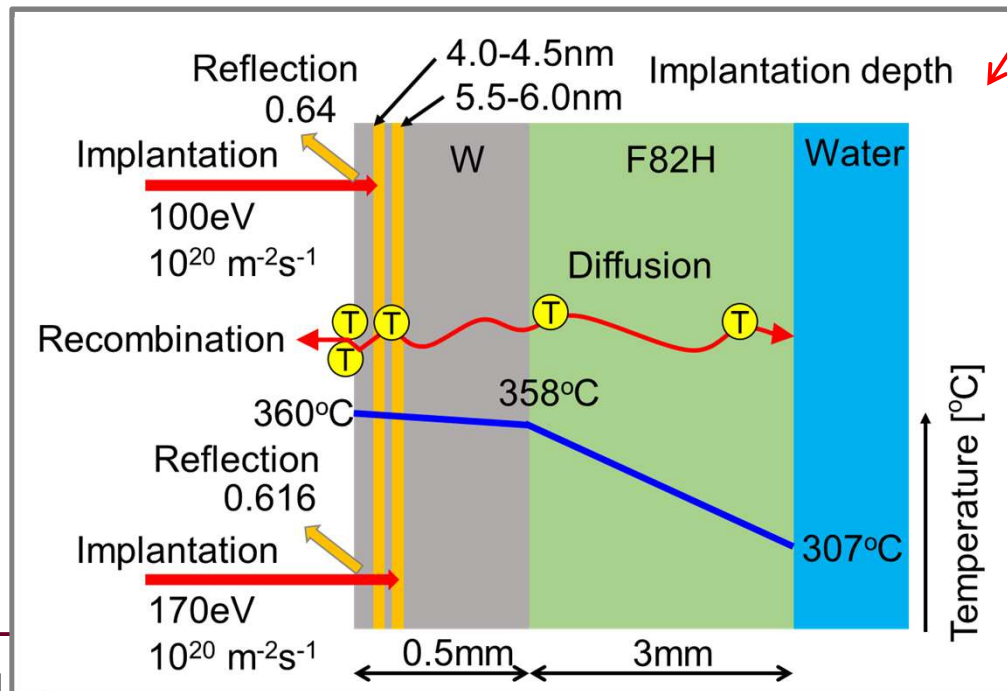


Exposed to divertor plasma

# Tritium permeation from plasma facing walls



JA-DEMO condition (tentative)



Tritium behavior was calculated by TMAP code.

Implantation depth was evaluated by SRIM code.

Tritium transport parameters were used from the data by Frauenfelder, Anderl for W, and Serra for F82H.

# T permeation rate to cooling water in the blanket

JA-DEMO condition (tentative)

| Region     | Surf(m <sup>2</sup> ) | Surf T(°C) | Particle (1/m <sup>2</sup> /s) | Particle E(eV) | Material |
|------------|-----------------------|------------|--------------------------------|----------------|----------|
| First wall | 1067                  | 360        | Ion : $10^{20}$                | 100            | W-F82H   |
|            |                       |            | Neutral : $10^{20}$            | 170            |          |

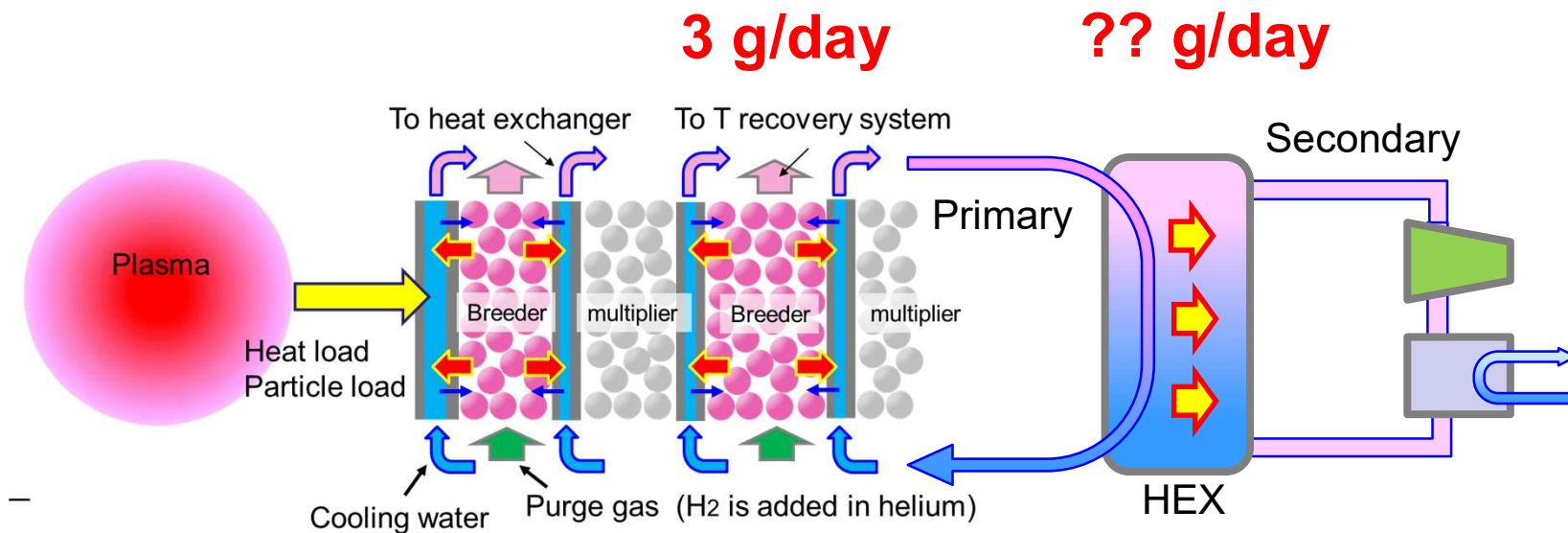
Evaluated permeation rate in the first wall to cooling water

| Region     | Steady state [hour] | Perm flux (g/m <sup>2</sup> /day) | Perm rate (g/day) |
|------------|---------------------|-----------------------------------|-------------------|
| First wall | 2                   | $6.49 \times 10^{-4}$             | <b>0.692</b>      |

Total permeation rate in the blanket modules

**2.3 g/day in breeding zone + 0.692 g/day in first wall  
= around 3 g/day**

# T permeation to secondary cooling water



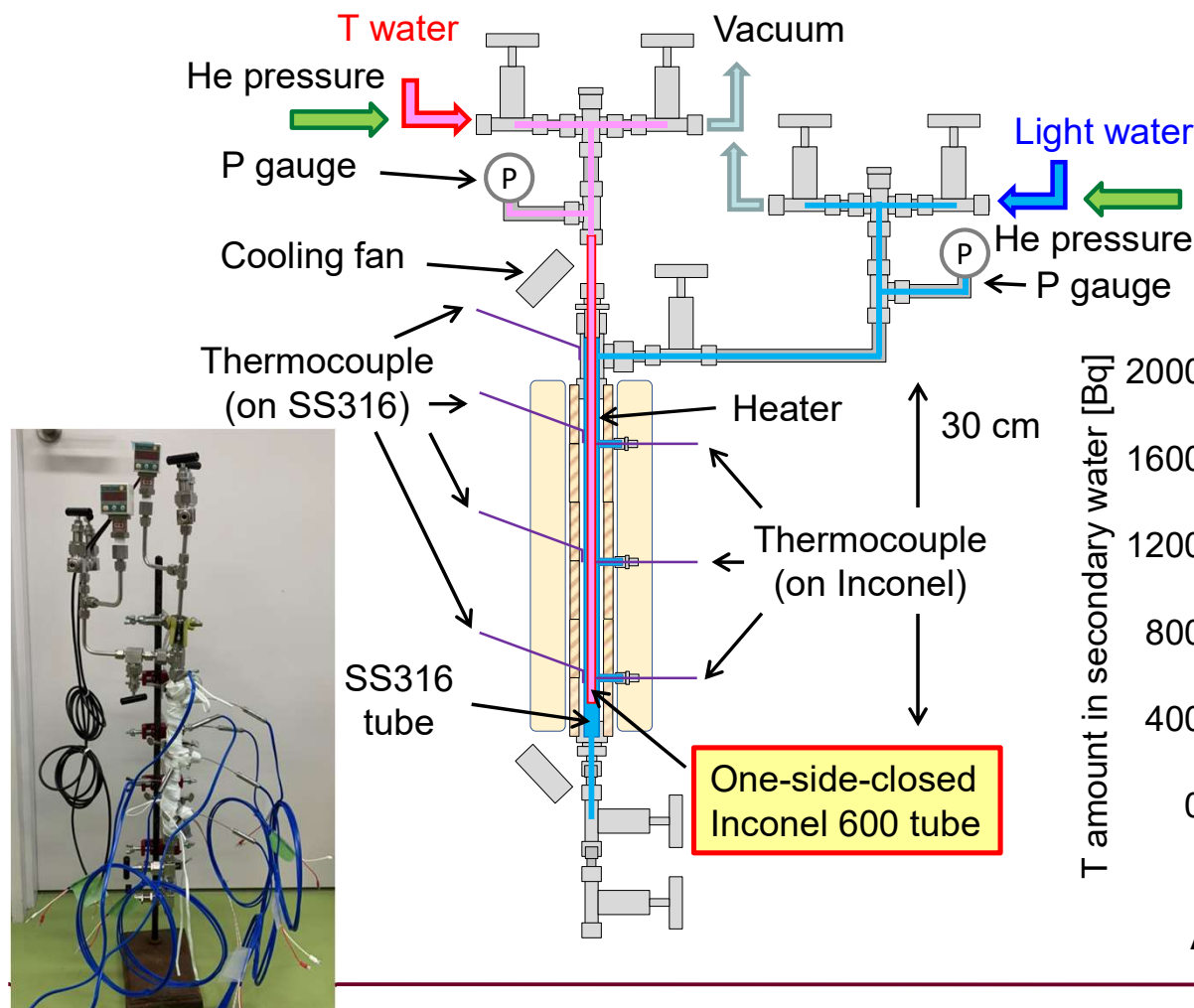
From the viewpoint of fusion safety and tritium management, it is necessary to understand **tritium permeation rate to the secondary cooling water**.

However, there are few reports on **tritium permeation between water and water** in heat exchangers. Therefore, we conducted a fundamental experiment and obtained initial data.



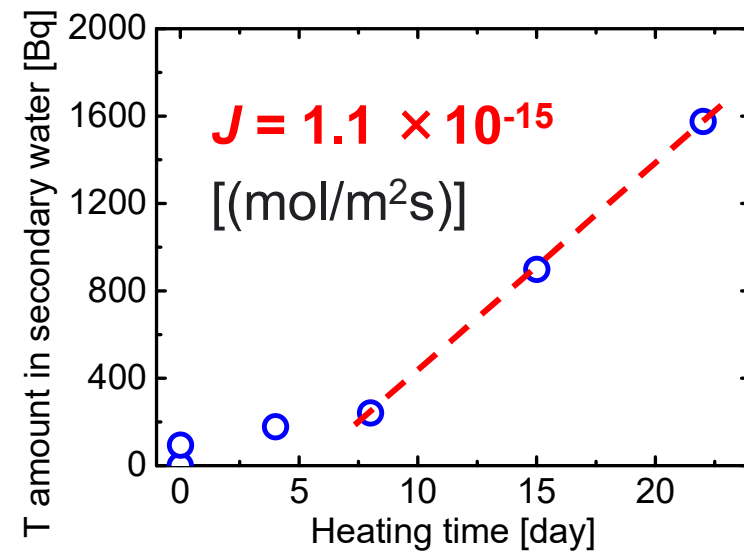
# Experiment of Water-Water T permeation

A double-tube permeation device consisting of **Inconel 600** (inner tube) and **SS316** (outer tube) was fabricated, and an experiment was conducted to observe T permeation between water and water through the Inconel 600 tube.



## Experimental conditions

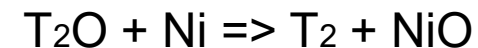
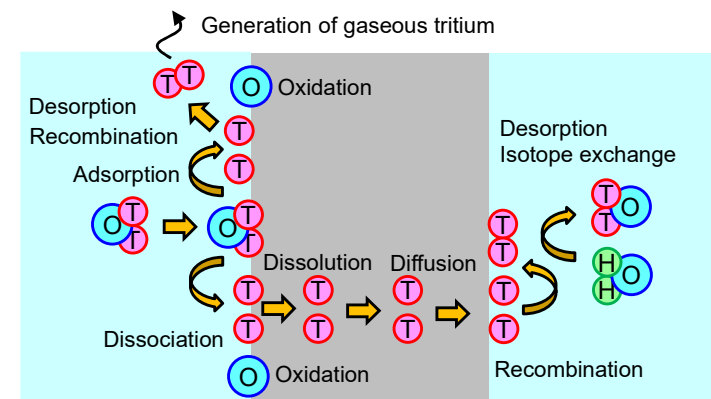
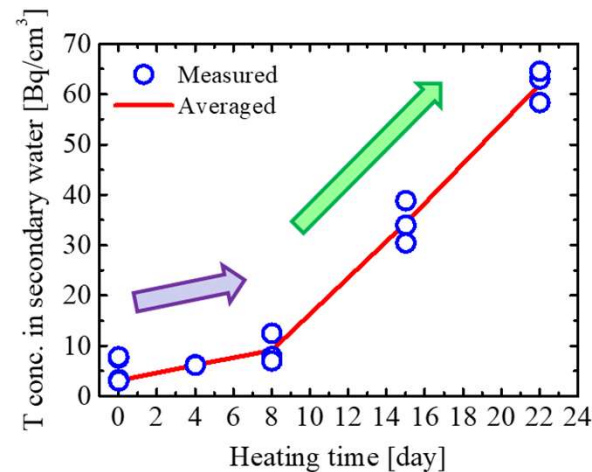
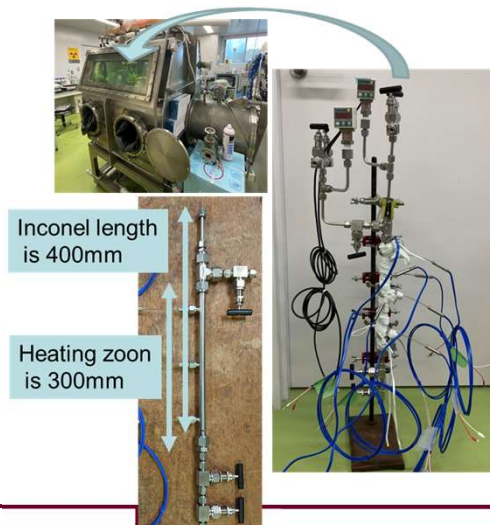
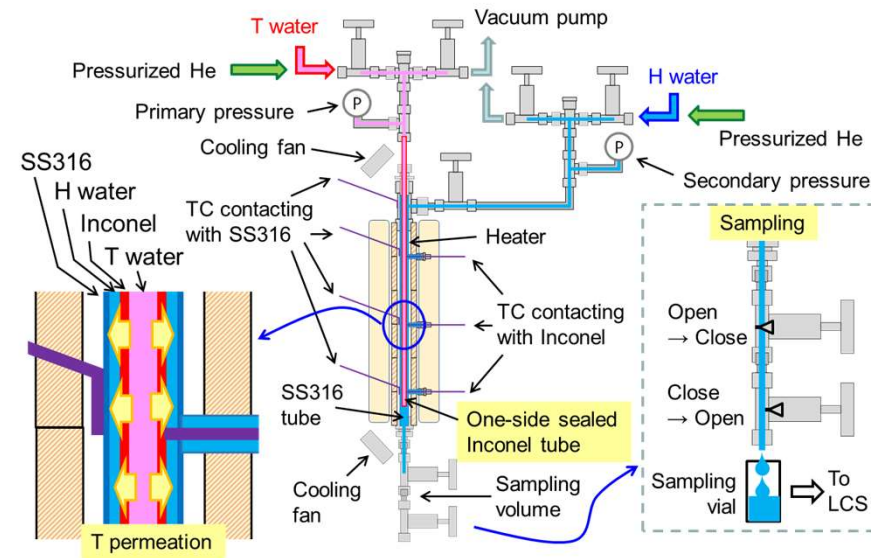
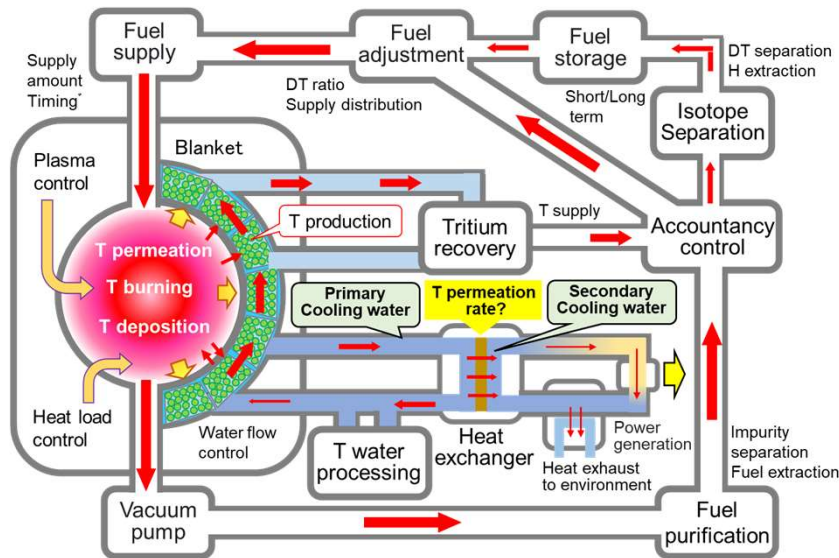
- T conc : 0.17 MBq/cc
- Pressure : 17 MPa
- Temp : 300 oC



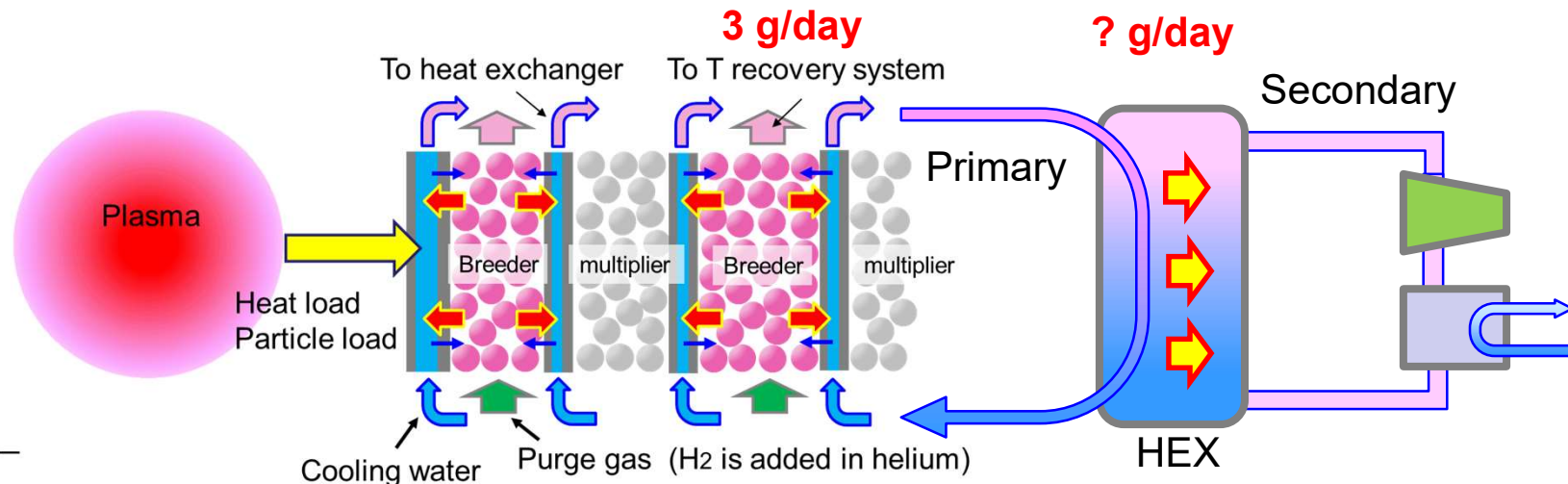
Amount of tritium permeated to secondary

# T permeation research from water to water

◆ From safety issue, we need to estimate T permeation rate to water.



# Evaluation of T permeation rate to Secondary



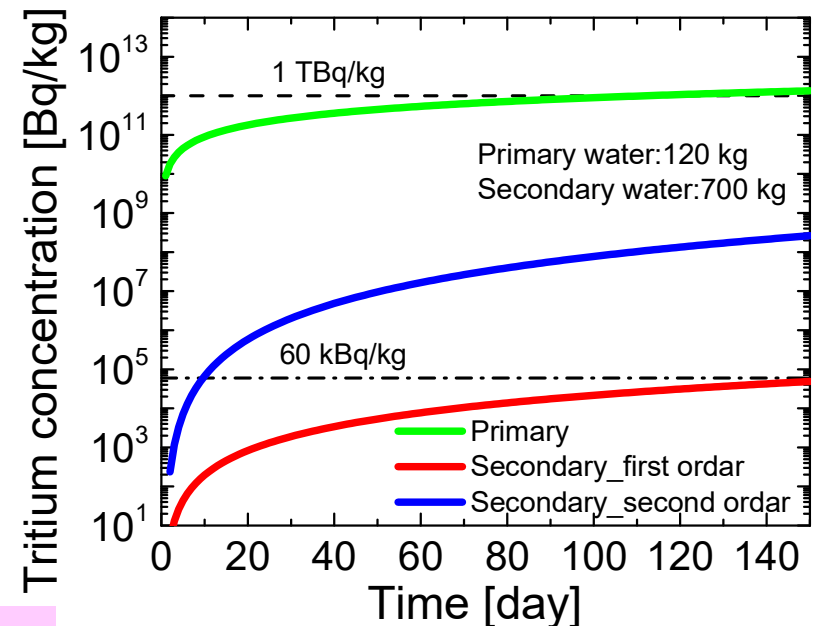
Open question : Dependence of permeation rate on T conc.

$$J \propto C_T \text{ or } J \propto C_T^2$$

Calculation conditions

- Material : Inconel 600
- Thickness : 0.5 mm
- Pressure : 17 MPa
- Temperature : 300 oC
- Permeation area : 23500 m<sup>2</sup>

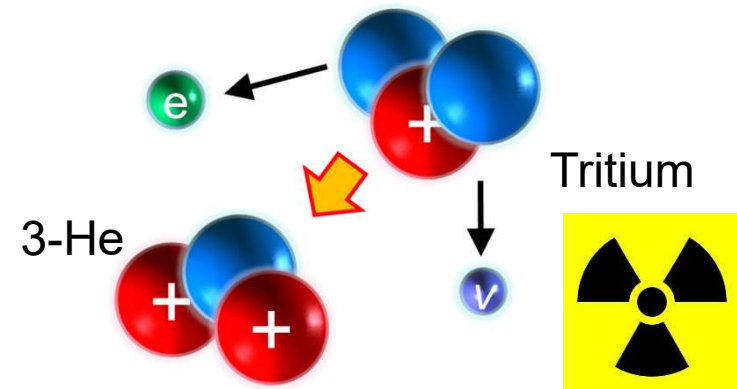
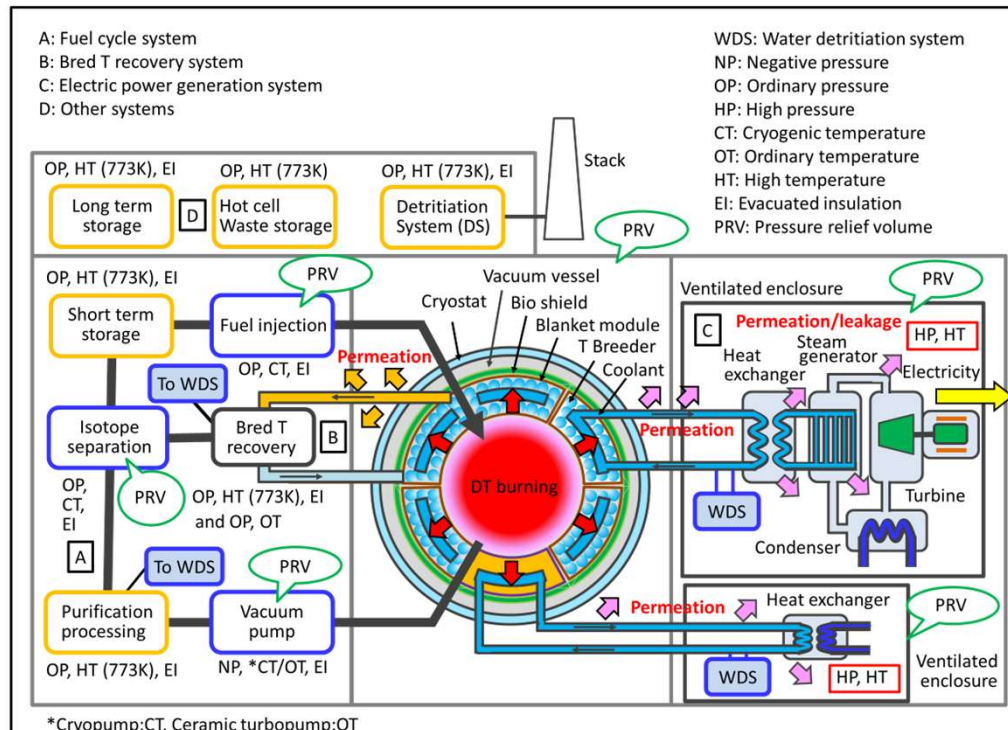
T concentration dependence on permeation rate strongly affects on evaluation of tritium conc in secondary.



Evaluation of T conc in secondary

# Tritium Chemical Plant

DT fusion plant is said to be “Tritium chemical plant”. In the fusion plant, tritium is nuclear burnt, produced, extracted, purified, isotope separated, injected.

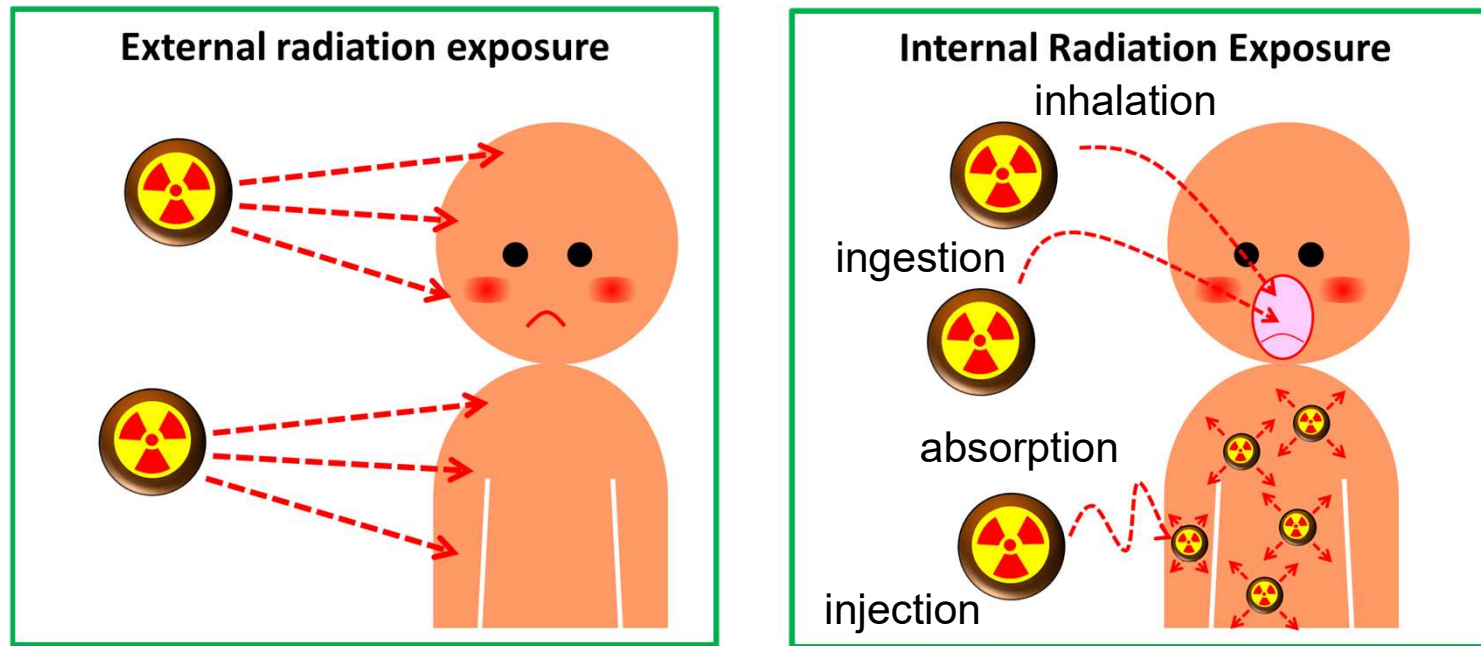


“Tritium” is radioactive isotope of hydrogen and it decays to  $^3\text{He}$  with a half life of 12.3 years in the process of beta decay.

- ◆ One important issue in the fusion engineering is the safety confinement of tritium in the plant.
- ◆ Another important issue in the fusion engineering is establishment of tritium cycle technology.



# Radiation protection from T



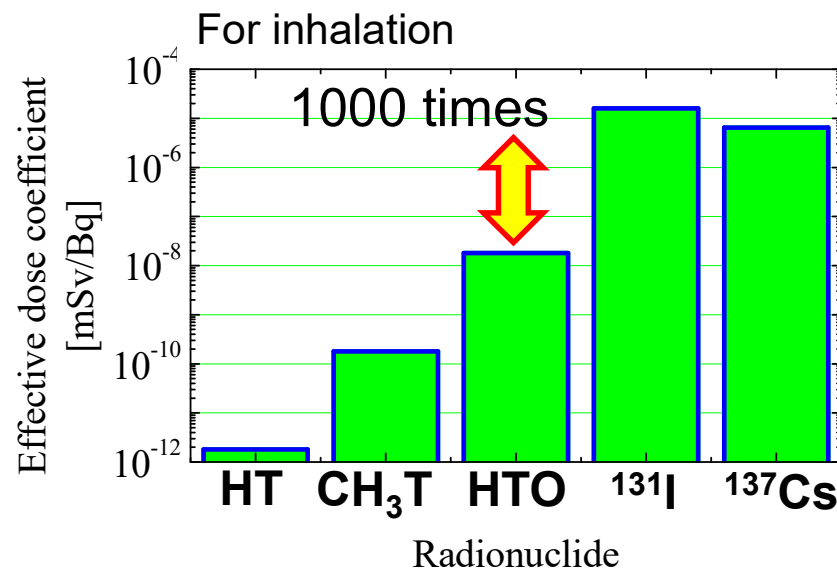
The maximum energy of  $\beta$  ray from T is 18.6 keV. The mean energy is 5.7 keV. The range of the  $\beta$  ray in tissue is less than 6  $\mu\text{m}$  but the thickness of skin is around 70  $\mu\text{m}$ . Therefore, the radiological effect by the external exposure of T is quite weak. Hence, the protection of **the internal exposure of T is mainly concerned**.

Tritium can enter the body through the pathway such as inhalation, ingestion, absorption and injection.

# Radiological impact from radioactive gas

Effective dose coefficient for inhalation of radioactive gasses

| Nuclide/chemical form                | Half-life | Coefficient [Sv/Bq]   |
|--------------------------------------|-----------|-----------------------|
| Tritium gas (T <sub>2</sub> )        | 12.3 y    | $1.8 \times 10^{-15}$ |
| Tritiated Methane (CT <sub>4</sub> ) |           | $1.8 \times 10^{-13}$ |
| Tritiated water (T <sub>2</sub> O)   |           | $1.8 \times 10^{-11}$ |
| I-131                                | 8.04 d    | $1.5 \times 10^{-8}$  |
| I-133                                | 20.8 h    | $3.1 \times 10^{-9}$  |
| Cs-137                               | 30 y      | $1.3 \times 10^{-8}$  |



Equivalent to 1 μSv exposure

|                   |         |
|-------------------|---------|
| <sup>131</sup> I  | 63Bq    |
| <sup>137</sup> Cs | 154Bq   |
| HTO               | 55600Bq |

**Radiological risk of tritium is not high.**

※<sup>131</sup>I : vapor, <sup>137</sup>Cs: all chemical component

# Points for handling T safely

In the safety operation of a fusion power plant, the most important issue is to confine T in the plant. Several large-scale T handling facilities have been constructed mainly to perform R&D for T handling in a fusion reactor and, various results and experiences for the T handling have been accumulated

## Tritium handling facility

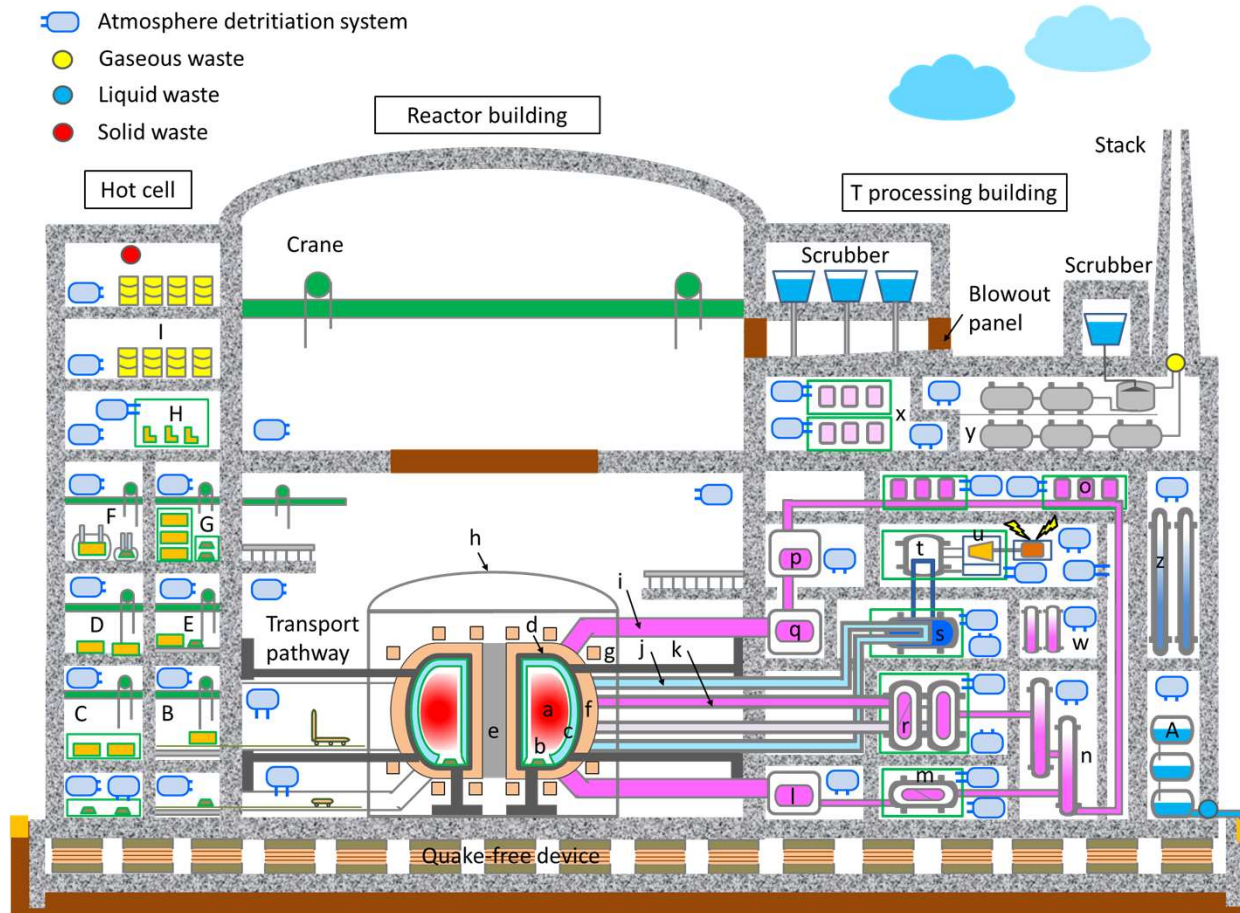
|                             | TPL<br>(Japan) | TSTA<br>(USA) | TLK<br>(Germany) | TFTR<br>(USA) | JET<br>(UK) | Kyushu UNIV. |
|-----------------------------|----------------|---------------|------------------|---------------|-------------|--------------|
| Allowed usage<br>amount [g] | 62             | 200           | 40               | 5             | 90          | 0.65mg       |

The following are fundamental and important points for handling T safely.

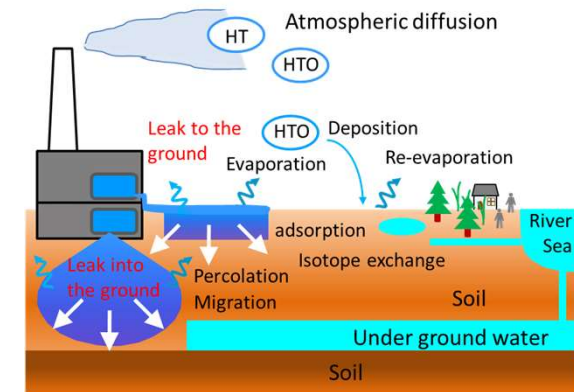
1. To equip multiple physical barriers.
2. To maintain the air flow from T low-concentration space toward T high concentration space.
3. To limit T inventory in any compartments.
4. To install T monitoring and detritiating systems in each compartment.
5. To design an appropriate detritiation system for each compartment considering the amount of T and its chemical form handled.
6. To avoid spreading T contamination.



# Tritium confinement and environmental impact

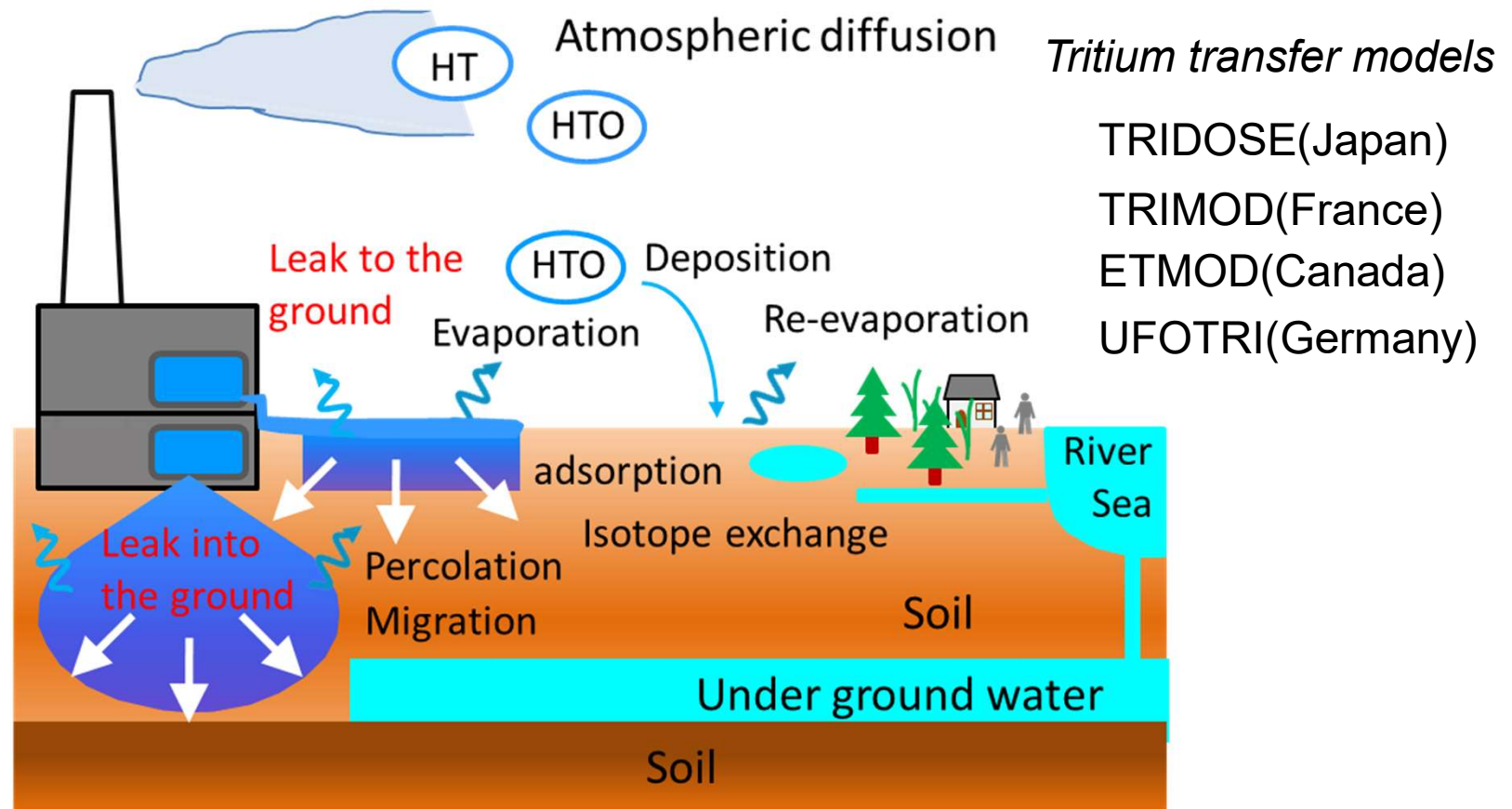


- |                          |                         |                      |                        |                      |
|--------------------------|-------------------------|----------------------|------------------------|----------------------|
| a: Plasma                | h: Cryostat             | o: T storage (short) | w: WDS (H level)       | D: Temporary storage |
| b: Divertor              | i: Fuel cycle pipe      | p: Fuel adjuster     | x: T storage (long)    | E: Maintenance       |
| c: Blanket               | j: coolant pipe         | q: Fuel injector     | y: Detritiation system | F: Inspection        |
| d: Vacuum vessel         | k: Bred T recovery pipe | r: Bred T recovery   | z: WDS (L level)       | G: Preparation       |
| e: Central solenoid coil | l: Vacuum pump          | s: Heat exchanger    | A: Water tank          | H: Waste management  |
| f: Toroidal coil         | m: Purification         | t: Steam generator   | B: Dust removal        | I: Waste storage     |
| g: Poroidal coil         | n: Isotope separation   | u: Turbine           | C: T removal           |                      |





# Tritium migration in natural soil

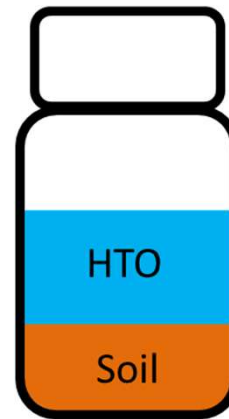


In a fusion reactor, a large amount of tritiated water is handled. It is necessary to assume a severe accident including release of tritiated water to the environment.

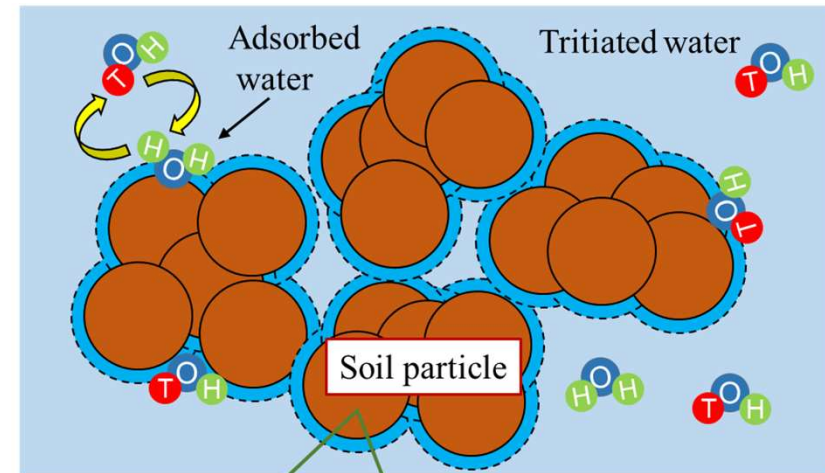
# Immersion of natural soil into tritiated water

Sample soil was immersed in tritiated water and tritium concentration in water was inperiodically measured

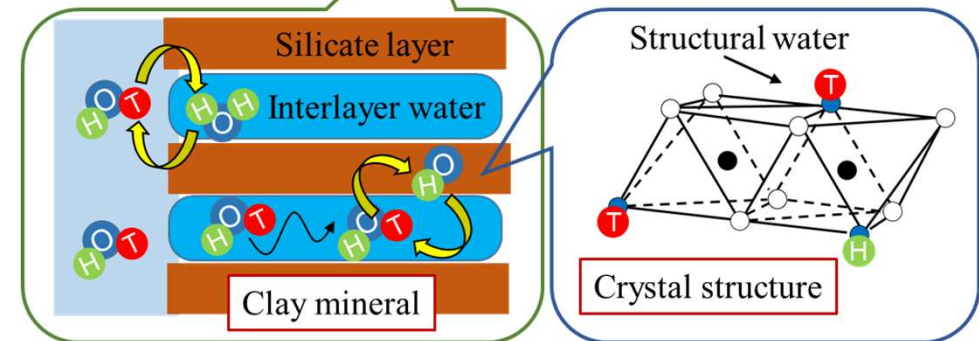
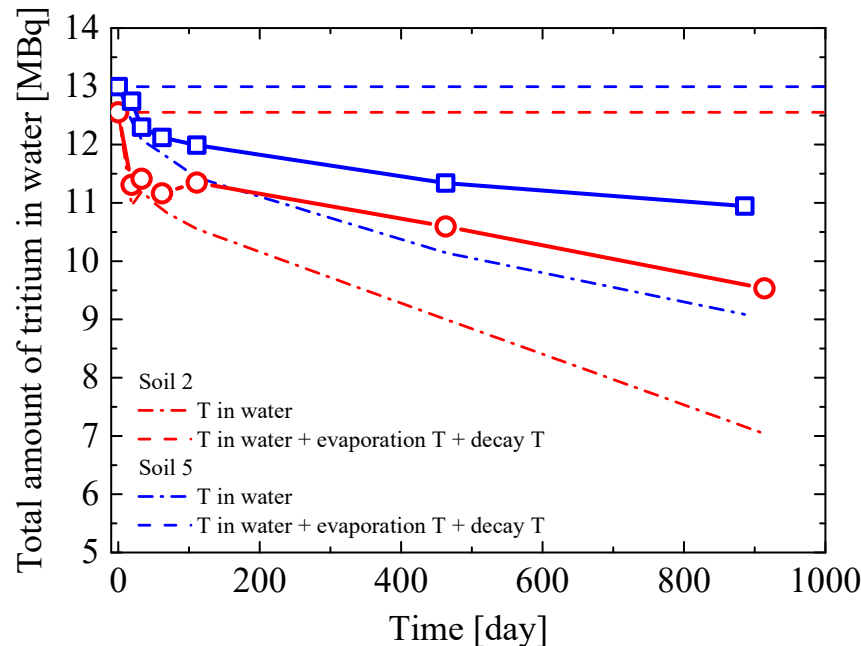
Tritium concentration decreased gradually with immersion time.



Teflon container



Tritium is trapped in soil.



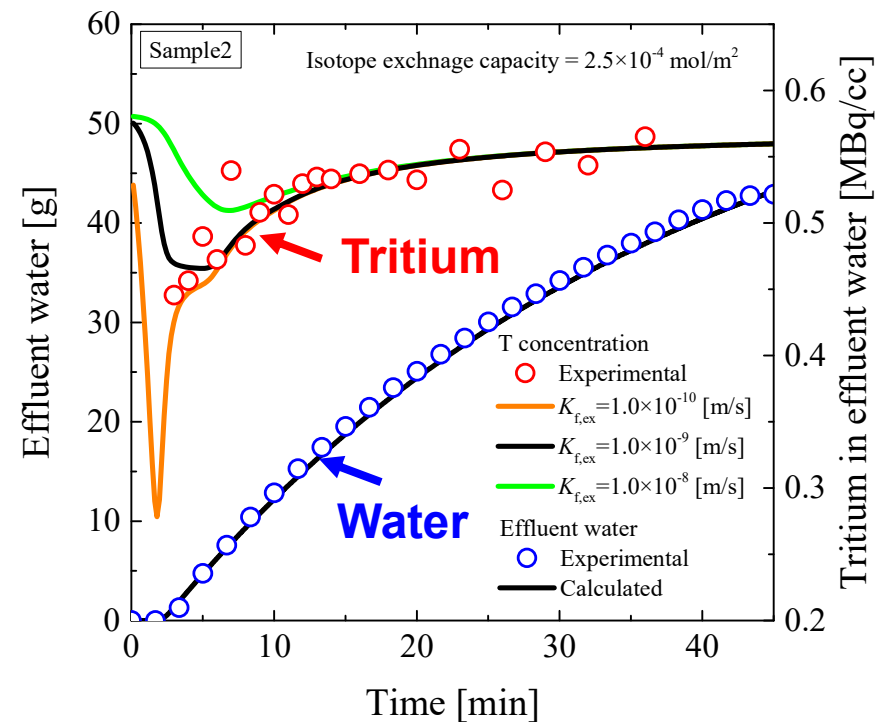
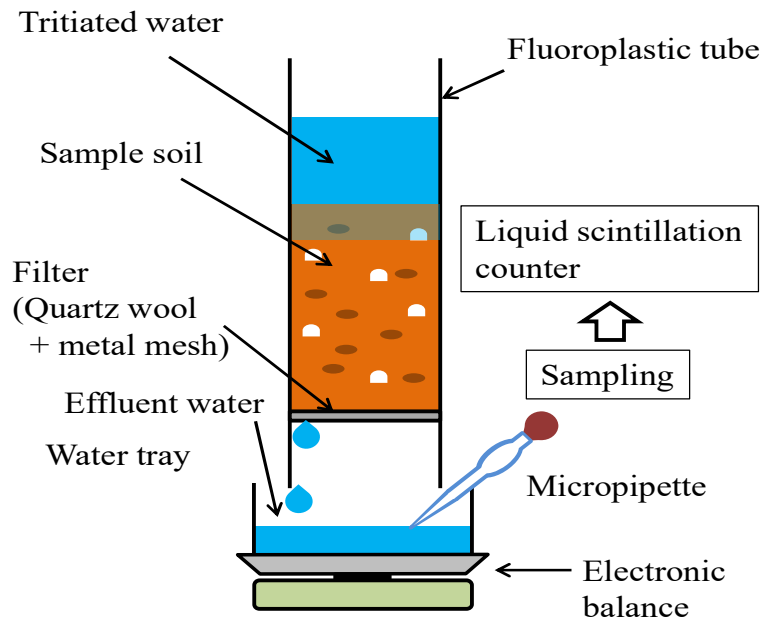
- Adsorbed water on the surface
- Interlayer water between silicate layers.
- Structural water such as OH-bases.

# Tritium migration in natural soil

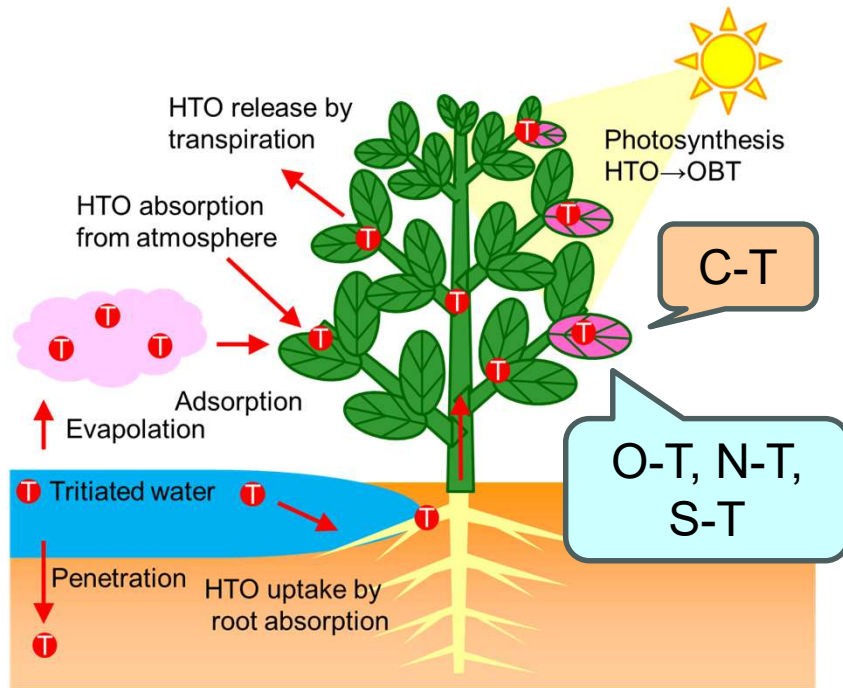


Sample soils were collected at 6 locations in the HAKOZAKI-Campus of KU.

Tritiated water was poured in the soil packed column and the weight of effluent water and its tritium concentration was measured.



# Transfer of T from soil to plants



Tritium in plants is categorized to two states.

Tissues free water tritium (TFWT)

(80-90%)<sup>[1]</sup>

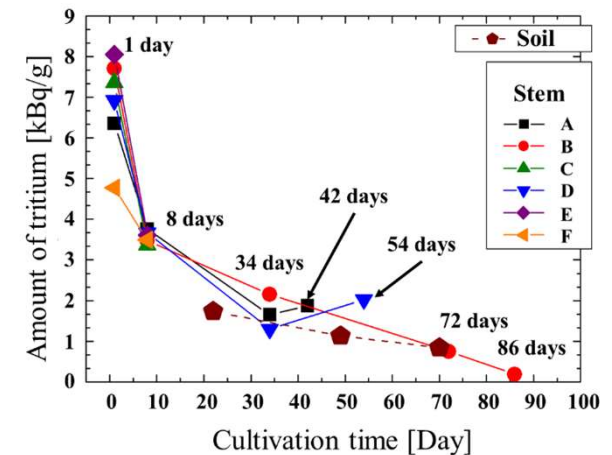
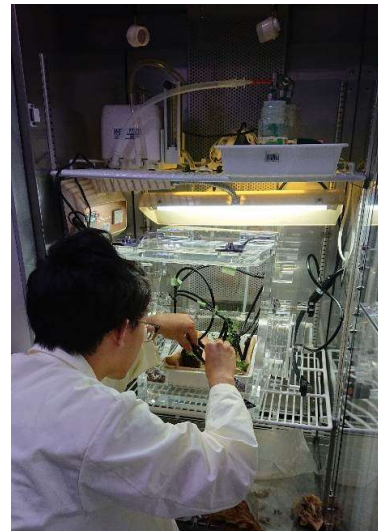
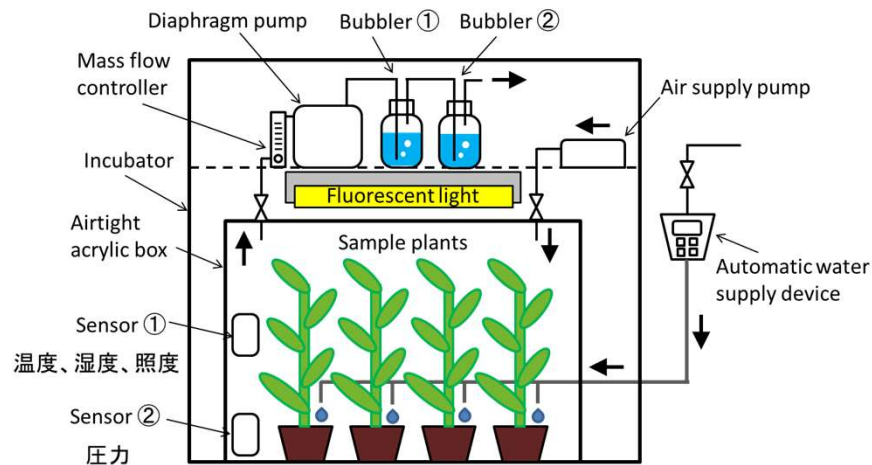
Organically bound tritium (OBT)

(5-10%)<sup>[1]</sup>

Isotope Exchangeable (E-OBT)

Isotope Non-exchangeable (Ne-OBT)

[1] C.Boyer, et al., Environmental and Experimental Botany 67 (2009) 34-51.



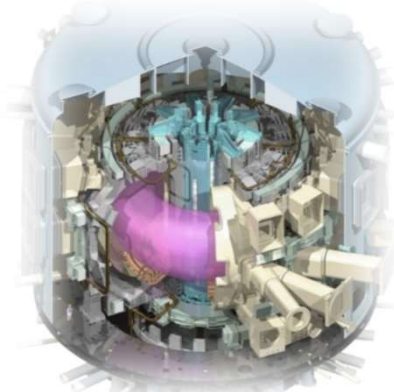
Tritium in plant and soil.



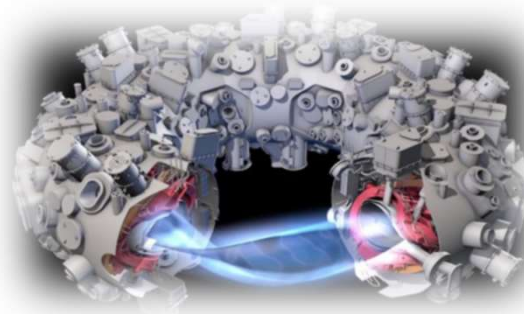
# Summary

Fusion reactors are attractive as next-generation energy sources because its fuel resources can collect from seawater.

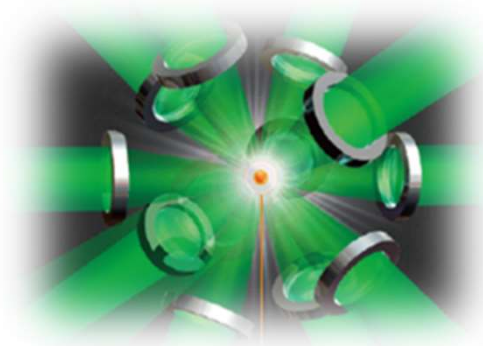
Several types of fusion reactors are under research and development worldwide.



Tokamak fusion reactor



Helical fusion reactor



Laser fusion reactor

In order to realize D-T fusion reactors, Tritium Science and Technology is a Key issue.

Fundamental technology for realization of D-T fusion reactors:

- ✓ Demonstration of fuel cycle using a large amount of tritium
- ✓ Development of tritium simulator for safety control
- ✓ Evaluation of radiological impact of tritium in environment.



# Report

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1. Describe the importance of tritium research in the development of fusion reactors.
2. Describe your thoughts on future energy sources after fossil fuels are exhausted.